FACULTY OF ENGINEERING

DEPARTMENT OF INDUSTRIAL ENGINEERING

SENIOR DESIGN PROJECT

FINAL REPORT:

Integrating Electric Buses to Istanbul Public Transportation System: Choosing the "right" routing and number of buses

Group Members:

Fatih Berker Akyıldız - 2019402162
Rahime Serra Topaloğlu - 2019402159
Ecem Öztürk - 2022402288
Ceren Yeralçak - 2022402297

Advisor: Taner Bilgiç

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Abstract

This project investigates the feasibility of converting diesel-powered bus lines into electric-powered alternatives in a pilot region, the Historical Peninsula in İstanbul, under different seasonal conditions and electric bus alternatives. Electric buses have many advantages over traditional diesel-powered buses such as reduced environmental and noise pollution. However, electric buses have limitations such as their limited travel distance and need for charging infrastructure. The energy consumption of an electric bus is affected greatly by passenger load, slope of the road, traffic distribution, and weather conditions. As a result, a route's feasibility for electrification requires careful examination.

A discrete-event simulation on SimPy was developed to assess the electrification feasibility of routes within the Historical Peninsula.The model uses real-world data including actual route lengths with slope percentages for each route segment and passenger distributions for each route provided by IETT. Based on predefined decision criteria, the simulation results indicated that only a subset of candidate routes could be electrified. This research provides a decision-support tool for IETT that can be utilized for extending electric bus deployment in broader regions of İstanbul.

Özet

Bu proje, farklı mevsim koşulları altında farklı elektrik otobüs çeşitleri kullanarak İstanbul Tarihi Yarımada'daki bir pilot bölgede dizel yakıtlı otobüsler ile çalışan otobüs hatlarının elektrikli otobüslerle çalışabilecek hale dönüştürülmesinin fizibilitesini araştırmaktadır. Elektrikli otobüsler, geleneksel dizel yakıtlı otobüslerle karşılaştırıldığında azaltılmış çevre ve gürültü kirliliği gibi birçok avantaja sahiptir. Ancak elektrikli otobüslerin sınırlı seyahat mesafesi ve şarj istasyonu altyapısına ihtiyaç duyması gibi sınırlamaları vardır. Elektrikli bir otobüsün enerji tüketimi yolcu yükü, otobüsün seyahat ettiği yolun eğimi, trafik dağılımı ve hava koşullarından büyük ölçüde etkilenir. Sonuç olarak bir güzergahın elektrikli otobüslerle çalışmasının fizibilitesi dikkatli bir inceleme gerektirir.

Tarihi Yarımada içindeki güzergahların elektriklendirme fizibilitesini değerlendirmek için SimPy üzerinde bir ayrık olay simülasyonu geliştirilmiştir. Model, her güzergah segmenti için eğim yüzdeleri ve İETT tarafından sağlanan her güzergah için yolcu dağılımlarını içeren gerçek dünya verilerini kullanmaktadır. Önceden tanımlanmış karar kriterlerine dayanarak, simülasyon sonuçları potansiyel güzergahların yalnızca bir alt kümesinin elektriklendirilebileceğini göstermiştir. Bu araştırma, İETT'ye İstanbul'un daha geniş bölgelerinde elektrikli otobüs kullanımının yaygınlaştırılmasında kullanılabilecek bir karar destek aracı sunmaktadır.

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1. Introduction

Istanbul's large population and complex urban structure create significant transportation challenges. Diesel-powered buses are the main component of road transportation, forming a significant part of the public transportation network. The high carbon emissions, rising fuel prices and maintenance needs of these vehicles are incompatible with both environmental and financial sustainability goals. Istanbul Electric Tram and Tunnel Operations (IETT) has strategically prioritized the integration of electric buses into the public transport network in line with its vision of zero emissions by 2050.

Electric buses (e-buses) provide both economic and environmental benefits with features like low maintenance costs, almost zero exhaust emissions, and low noise levels. However, there are a number of operational and technological limitations associated with this change. The system integration process necessitates thorough planning and research due to the e-buses' limited range, reliance on charging infrastructure, and high investment cost per vehicle.

In accordance with the 2050 sustainability vision set up by IETT, a decision support model was created for the incorporation of electric buses into the Istanbul public transportation system, addressing multi-criteria decision-making problems through simulation modeling. By identifying the best routes and necessary fleet size based on specific parameters, the study aims to maximize the deployment of e-buses in terms of operational efficiency, passenger demand, and environmental impact.

The performance of several bus models on different routes was assessed in this context through analytical and simulation experiments conducted on a particular pilot region. Due to factors including heavy passenger traffic, the density of short-distance lines, and ongoing electric bus trials, the area chosen as the pilot region offers a sufficient sample for research.

1.1 Background: Urban Mobility Challenges and Decarbonization Strategies in Istanbul

Istanbul's dense population, rising usage of private vehicles, and inadequate physical infrastructure capacity provide significant operational and environmental challenges for the city's transportation system. As of 2022, the significant portion of Istanbul's urban carbon emissions is attributed to the transportation sector. Accordingly, the Istanbul Metropolitan Municipality (IBB) aims to establish a carbon-neutral transportation system by 2050 through multi-stakeholder strategic planning. The public transportation system's reorganization has been identified as one of the plans' primary goals. Due to rising operational expenses and significant carbon emissions, IETT's fleet of diesel-powered buses is in the forefront of this change.

The integration of electric vehicles into the system is the special focus of the IBB-supported projects "Decarbonization of the Public Transport Bus Fleet" and "Decarbonization of the Metrobus." The adoption of low-emission technologies, improved energy efficiency, and sustainable urban mobility are all enabled by these initiatives.

The initial instance of the transition to electric buses was implemented on the 28T line between Topkapı and Beşiktaş. This line's vehicle is a diesel-to-electric bus conversion that is charged overnight and operates throughout the day on scheduled six trips. This pilot application allows examination of parameters including operational suitability, passenger experience, and energy consumption under real city conditions.

To ensure that this shift extends beyond vehicle technology, IBB has also put demand management measures like "Low Emission Zones", locally referred to as Düşük Salım Bölgeleri (DSB), into place. The foundation of the DSB application is the idea of prohibiting or taxing cars that don't adhere to emission regulations to enter specific zones.

This approach aims to promote the use of public transit and micromobility vehicles while decreasing the use of private vehicles in one of the most crowded parts of the city. In

Istanbul, the Historical Peninsula (Eminönü) was chosen as the first DSB pilot location. Later on, the application is expected to be expanded to areas such as Kadıköy. (Istanbul Metropolitan Municipality, 2022)

To offer comprehensive solutions for urban transportation, in addition to the DSB policy, complementing policies are implemented, such as Feeder Bicycle Routes projects and applications for electric bicycles and scooters. By facilitating first and the final kilometers travel, these technologies seek to improve the efficiency and accessibility of public transit networks.

1.2 Short Overview of Identification, Analysis and Solution Methodologies

In this project, the feasibility of converting diesel bus lines operating in the Historical Peninsula region of Istanbul to electric buses was investigated. The study is based on a systematic methodology that includes data collection, simulation modeling and route evaluation steps.

In the first stage, a comprehensive data analysis was performed using real-time passenger distribution, stop coordinates and line information obtained from IETT. In addition, slope profiles and distance data of the lines were collected via Google Maps API. With this data, discrete event simulation was developed using the SimPy library, and the performances of different electric bus models (e.g. BYD K8 and Karsan e-Atak) and diesel buses were compared. By integrating seasonal variables, passenger density and traffic effects into the simulation, more realistic results were obtained.

According to the simulation results, performance metrics such as energy consumption, CO_2 emissions, minimum battery level and trip completion rates were evaluated. These data were statistically analyzed with sample means and 95% confidence intervals to test their reliability. Finally, the lines were evaluated according to criteria such as operational suitability and environmental impacts, and the priority lines for conversion and the lines that could be addressed in the second stage were determined.

This applied methodology provides a data-based decision support mechanism on which lines electric buses can be used more efficiently and contributes to the development of sustainable transportation policies.

1.3 Improvements due to the Project

The current project deliberately limited its scope to bus routes operating within Fatih, a central and dense urban area. Fatih provided an ideal testbed not only because of its manageable distances, moderate slopes, and high passenger volumes also due to the vision of IETT. However, the city-wide deployment of electric buses will require analysis across much more diverse regions, including Kadıköy, Üsküdar, Beşiktaş, Eyüpsultan, and Beylikdüzü, where topographical, infrastructural, and route-length characteristics differ significantly. Importantly, routes outside Fatih may include long-haul and inter-district lines, which could challenge the operational viability of lithium-ion buses used in this study. In such cases, alternative technologies—such as higher capacity LFP batteries or next-generation solid-state batteries—can be evaluated for compatibility with extended-range operations. A comparative analysis of battery chemistries and vehicle types would help guide procurement and infrastructure planning decisions.

Another important future enhancement involves improving the temporal accuracy of the simulation by integrating real-time or historical traffic data. In the current study, average speeds for time blocks (morning, afternoon, evening, night) were used to represent traffic dynamics. While this was sufficient for a baseline model, real-world traffic in Istanbul is subject to extreme fluctuation due to accidents, weather, public events, and daily variability.

Finally, to support more strategic and long-term planning, future iterations of the simulation can incorporate battery degradation models. Battery performance, capacity, and efficiency degrade over time depending on factors like charging cycles, real-time operating temperature, and depth of discharge.

Incorporating degradation models—such as those proposed by Berecibar et al. (2016) and Dubarry et al. (2011)—would allow for the estimation of:

- Long-term replacement schedules for vehicle batteries,
- Lifetime cost of ownership comparisons between diesel and electric buses,
- Required changes in charging infrastructure capacity over time.

Integrating lifecycle effects into the model would shift the simulation from a short-term feasibility tool to a full-scale strategic planning instrument, offering insights into maintenance, budgeting, and long-term emissions.

This study considers only depot charging since IETT wants to deploy charging stations into the Topkapı Bus Depot. In future studies, alternative charging strategies -such as flash charging and opportunity charging- can be considered to evaluate their impact on route-level operational efficiency compared to depot charging. For instance, implementing the flash charging strategy would allow a higher number of routes to be feasible for electric bus deployment since it reduces the overall delays caused by depot visits for charging and the need for large battery sizes in buses. However, since this strategy comes with high installation and infrastructure costs, consideration is needed when budget constraints are a critical decision factor in future planning.

Although the investment cost for electric bus deployment is a major constraint in real-world applications, it was not a decision variable in this study. Should the budget constraints become critical in future studies, it is recommended to use an approach based on cost-effectiveness by adding cost per kg of CO_2 emissions as a decision variable before deciding the routes for electric bus deployment.

1.4 Summary of Conclusions

Within the scope of this study, a route-based evaluation model was developed for the integration of electric buses into the Istanbul public transportation system. The Historical Peninsula region was selected as a pilot area; simulation-based analyses were conducted by considering different bus types, passenger density and seasonal conditions. The findings show that the lines selected as the first (93, 33, 36KE, 35) and second (336E, 38E, 33B) phases are operationally suitable for electric bus conversion and that the conversion can significantly reduce environmental impacts. The developed model provides data-based support to decision makers in fleet planning and route selection decisions.

2. Problem Definition, Requirements and Limitations

2.1 Problem Definition

Istanbul's public transportation fleet consists of diesel-powered buses, which are an important contributor to the city's air pollution and carbon emissions. An economically unsustainable structure is additionally being established by rising fuel prices and maintenance expenses. In this regard, switching to electric buses for the IETT fleet in the aim of meeting its zero-emission goal by 2050 presents a number of difficulties that require careful infrastructural and operational management.

E-buses differ significantly from diesel buses in terms of range restrictions, charging schedules and energy consumption profiles. Route planning and frequency are directly affected by the fact that electric buses can typically only be fuelled at slow or medium-speed charging stations located in garages, whereas diesel buses offer more flexibility due to their quick refueling.

Furthermore, the structure of Istanbul's current bus lines is highly varied with regard to route length, traffic conditions, topographical slope, and passenger density. As a result, implementing a single electric bus distribution plan that works for each route is not feasible. Line-based assessments, real-world vehicle performance evaluations, and fleet planning are all required for the efficient use of electric buses.

There currently exists no integrated decision support tool in the IETT system that would allow for the line-by-line assessment of electric buses' operational suitability and environmental effects. There are potential risks associated with this scenario that could lead to inefficient investments and interruptions in the quality of services.

Finding the routes and fleet size where the deployment of electric buses will have the most impact on operational efficiency and environmental advantages is the study's primary objective in this regard. A simulation-based analysis approach has been created for this aim, taking into account technical characteristics including slope, passenger density, traffic conditions, seasonal circumstances, and energy consumption.

2.2 Needs and Requirements of System

At the beginning of the project, meetings were held with IETT’s R&D and Demand Planning teams in order to better analyze the source of the problems in the current transportation system. As a result of these meetings, comprehensive information was obtained about the operational structure of electric buses, charging times, range limits and technical details that should be taken into account in fleet planning. In addition, data such as passenger density, stop structure and frequency of trips for diesel bus lines were shared and the current status of the system was clearly revealed.

In the meetings with IETT, it was stated that the main goal of the institution is to reach the zero emission vision by 2050. In this context, it was emphasized that electric bus projects that will reduce CO_2 emissions and contribute to environmental sustainability are supported. The institution requested that an analysis be conducted on a line basis, operational suitability criteria be evaluated and decisions be supported with a prioritization system in order for this transition to proceed in a planned manner.

The Historical Peninsula was selected as the pilot area. The area is strategically important due to its high passenger density, historical and environmental sensitivity, and hosting the first Low Emission Zone (LSB) application implemented in Istanbul. In this context, the main purpose of the study was defined as determining the most suitable lines and required fleet sizes where electric buses can be applied in a way that will reduce emissions while maintaining service quality.

2.3 Limitations and Constraints

The simulation model developed in this study was designed within the framework of various limitations and data constraints. Firstly, legal regulations such as weight limits determined for buses create a limit on the choice of buses and the number of passengers that can be carried, and therefore finding the type of buses that can be used in the model was a challenge. Since there is no real stop-based data on passenger numbers, boardings are distributed uniformly along the route. This assumption may cause bus occupancy rates and passenger comfort levels to appear higher than they actually are. For instance, a 60% utilization rate was calculated in the model; however, this may not reflect the actual hourly and stop-based variability. The lack of this data was a challenging situation when developing the model.

Furthermore, air conditioning usage, which varies depending on climatic conditions, is modeled with the assumption of constant temperature value throughout the season. This may cause energy consumption estimates to deviate from real environments. In the carbon emission calculations, the average electricity-related emission value of 0.233 kg CO_2/kWh was used. However, this value does not take into account regional and temporal changes in electricity generation methods.

All these limitations affect the accuracy of the simulation, and the realism of the model can be increased by using higher resolution data sets and dynamic variables in future studies.

2.4 Data Gathered and Used in the Identification Phase

The data sources used in this study are categorized into three groups which are the data provided directly by IETT, data obtained through open APIs, and information gathered through external research.

2.4.1 Data Provided by IETT

Since the Topkapı Garage was prioritized by IETT and the Historical Peninsula was determined as the pilot area, the focus was on bus lines using the Topkapı Garage and having Eminönü as their last stop. The dataset provided by IETT included information such as the stops of all lines in Istanbul, the coordinate information of the stops, the hourly number of passengers on the lines, the number of daily trips, and the number of vehicles used on the line. Different filtering and prioritizations were made to examine this dataset.

Firstly, in order to find lines suitable for the scope of the study, the bus lines that use the Topkapı garage and have the Eminönü last stop were filtered and the coordinate information of the stops to be used for the analysis was found.

The dataset includes the stops that each line passes through in order and the geographical coordinates of these stops (Table 1). The original Excel file in the dataset contained 11,395 rows and 12 columns. When only the lines and related variables within the scope of the analysis were selected, the data was reduced to 982 rows and 8 columns.

Table 1- *Screenshot of the stops dataset*

Line Code	Route	Stop Name	Stop Code	Stop Type	X	Y	Order
28T	28T_D_D0	BEŞİKTAŞ MEYDAN	114991	WALLMODERN	29,006855	41,043615	1
28T	28T_D_D0	KEMERALTI	117411	İETTBAYRAK	28,976892	41,024981	8
28T	28T_G_D0	TOPKAPI	105552	CCMODERN	28,924866	41,020602	1
28T	28T_G_D0	HASEKİ	105453	CCMODERN	28,944259	41,010524	8
28	28_D_D0	BEŞİKTAŞ MEYDAN	114991	WALLMODERN	29,006855	41,043615	1
28	28_D_D0	KEMERALTI	117411	İETTBAYRAK	28,976892	41,024981	8
28	28_G_D0	TOPKAPI	105552	CCMODERN	28,924866	41,020602	1
28	28_G_D0	YAVUZ SELİM	105791	WALLMODERN	28,945726	41,021239	8
32	32_D_D0	EMİNÖNÜ	308031	WALLMODERN	28,965788	41,019795	1
32	32_D_D0	ŞEHİT YUNUS EMRE EZER	105822	WALLMODERN	28,938748	41,027383	8
32	32_G_D0	CEVATPAŞA	301141	CCMODERN	28,883111	41,072175	1
32	32_G_D0	YILDIRIM LUNAPARK	106611	WALLMODERN	28,900538	41,060188	8
33	33_D_D0	EMİNÖNÜ	311032	İETTBAYRAK	28,965343	41,02012	1
33	33_D_D0	FINDIKZADE	105692	WALLMODERN	28,940261	41,012149	8
33	33_G_D0	GİYİMKENT	128531	İETTBAYRAK	28,871979	41,074212	1
33	33_G_D0	YÜZYIL-ORUÇREİS METRO	185391	İETTBAYRAK	28,85564949	41,06320243	8
34	34_D_D0	ZİNCİRLİKUYU	900081	ISTASYON	29,01436161	41,06623281	1
34	34_D_D0	AYVANSARAY EYÜPSULTAN	900151	ISTASYON	28,938211	41,039265	8
34	34_G_D0	AVCILAR	900333	ISTASYON	28,73079255	40,98602508	1
34	34_G_D0	BEŞYOL	900272	ISTASYON	28,79466594	40,99370114	8
35	35_D_D0	EMİNÖNÜ	307031	WALLMODERN	28,966094	41,019467	1
35	35_D_D0	HASEKİ	105452	İETTBAYRAK	28,944059	41,007788	8

The columns used contain the following information:

- **Line code:** This column shows the bus route details such as 28T, 28 etc.
- **Route:** This column shows the direction with the specific line codes, D represents return, G represents departure.
- **Stop Name:** This column shows the name of the bus stop and is unique for each stop.
- **Stop Code:** This column shows the code of the bus stop and is unique for each stop.
- **Stop Type:** This column indicates the physical type of the stop such as IETT flagged stop, modern glass stall etc.

- **X:** This column shows the longitude coordinate of the stop.
- **Y:** This column shows the latitude coordinate of the stop.
- **Order:** This column shows the order of the stop on the route.

The dataset provided by IETT also included the number of vehicles and daily trips based on the line. Using this information, the average daily trips per vehicle was calculated. Here, it was assumed that each bus made the same number of trips. As seen in Table 2, lines with an average trip number of more than the average were marked as frequent. This information will be used when evaluating the lines at the end of the simulation.

Table 2- *Screenshot of the schedule dataset*

Line Code	# of Vehicles	# of Bus Trips	# of Trips per Vehicle	Schedule Type
32	7	72	10	Frequent
33	7	56	8	Inrequent
35	13	172	13	Frequent
80	1	4	4	Inrequent
90	4	88	22	Frequent
93	9	90	10	Inrequent
28T	8	92	12	Frequent
336E	25	248	10	Inrequent
33B	9	89	10	Inrequent
33E	1	3	3	Inrequent
33ES	2	9	5	Inrequent
33TE	3	15	5	Inrequent
33Y	3	41	14	Frequent
36KE	6	52	9	Inrequent
38E	8	90	11	Frequent
99A	10	172	17	Frequent
		Avg:	**10**	

The columns used contain the following information:

- **Line code:** This column shows the bus route details such as 28T, 28 etc.
- **# of Vehicles:** This column shows the total number of buses used in that line.
- **# of Bus Trips:** This column shows the total number trips in that line.
- **# of Trips per Vehicle:** This column shows the total number trips per vehicle in that line with the equal number of trips assumption.
- **Schedule Type:** This column shows if the schedule of that line is frequent or not.

The dataset provided by IETT also includes hourly passenger boarding information for each line. This data for the lines on the European Side was taken from two different periods (January 2025 and July 2024) and raw data is shown in Table 3. In this way, demand profiles specific to the summer and winter months could be analyzed separately. These datasets consist of 802,384 rows for the winter period and 770,449 rows for the summer period. In the pre-processing phase, the data was cleaned by eliminating lines not included in the analysis and unnecessary variables. Then, the average, maximum and minimum passenger numbers were calculated for each bus line.

In order to reflect the demand changes more accurately during the day, time was analyzed in four blocks: morning and evening were defined as rush hours, and noon and night were defined as normal hours. For each season and time block, maximum and minimum passenger numbers were determined on a bus line basis.

Table 3- *Screenshot of the raw passenger dataset for January 2025*

Güzergah Hat Kodu	Araç Uzunl	Operatör G	Tarih	Araç Kapı	Güzergah	Servis No	Güzergah Yön	Planlanan Basl	Saat Tekil	Görev Dur	Sefer Bazlı Yolculu ayısı
303A	11-14	ÖHO	01.01.2025	A-902	303A_G_D0	-1	GİDİŞ	04:00:00	4	T	20
303A	11-14	ÖHO	01.01.2025	A-903	303A_G_D0	-3	GİDİŞ	04:15:00	4	T	2
303A	11-14	ÖHO	01.01.2025	A-904	303A_G_D0	-5	GİDİŞ	04:30:00	4	T	10
401	6,5-8	ÖHO	01.01.2025	A-1051	401_G_D0	-1	GİDİŞ	04:30:00	4	T	9
146	14-19	İETT	01.01.2025	K1559	146_G_D7745	3605	GİDİŞ	04:40:00	4	T	80
401	6,5-8	ÖHO	01.01.2025	A-1052	401_G_D0	-3	GİDİŞ	04:40:00	4	T	1
418	6,5-8	ÖHO	01.01.2025	A-1218	418_G_D0	-1	GİDİŞ	04:40:00	4	T	13
303A	11-14	ÖHO	01.01.2025	A-905	303A_G_D0	-7	GİDİŞ	04:45:00	4	Z	0
401	6,5-8	ÖHO	01.01.2025	A-1053	401_G_D0	-5	GİDİŞ	04:50:00	4	T	4
418	8-9	ÖHO	01.01.2025	A-830	418_G_D0	-3	GİDİŞ	04:55:00	4	T	7
303A	11-14	ÖHO	01.01.2025	A-906	303A_G_D0	-9	GİDİŞ	05:00:00	5	T	33
401	6,5-8	ÖHO	01.01.2025	A-1054	401_G_D0	-7	GİDİŞ	05:00:00	5	T	2
448	6,5-8	ÖHO	01.01.2025	A-743	448_G_D0	-1	GİDİŞ	05:00:00	5	T	9
458	6,5-8	ÖHO	01.01.2025	A-667	458_G_D0	-1	GİDİŞ	05:00:00	5	T	27
89C	11-14	İETT	01.01.2025	B3001	89C_G_D3495	8931	GİDİŞ	05:00:00	5	T	53
TM17	11-14	ÖHO	01.01.2025	A-1021	TM17_G_D397	-13	GİDİŞ	05:00:00	5	T	27
401	6,5-8	ÖHO	01.01.2025	A-1065	401_G_D0	-9	GİDİŞ	05:10:00	5	T	5
418	6,5-8	ÖHO	01.01.2025	A-710	418_G_D0	-5	GİDİŞ	05:10:00	5	T	15
H-3	11-14	İETT	01.01.2025	B5261	H-3_G_D4224	4947	GİDİŞ	05:10:00	5	T	1
303A	11-14	ÖHO	01.01.2025	A-907	303A_G_D0	-11	GİDİŞ	05:15:00	5	T	21
429A	6,5-8	ÖHO	01.01.2025	A-681	429A_G_D0	-1	GİDİŞ	05:15:00	5	T	56
448	6,5-8	ÖHO	01.01.2025	A-744	448_G_D0	-3	GİDİŞ	05:15:00	5	T	3
458	6,5-8	ÖHO	01.01.2025	A-833	458_G_D0	-3	GİDİŞ	05:15:00	5	T	18
98	11-14	KOOP	01.01.2025	A-1682	98_G_D7742	-501	GİDİŞ	05:15:00	5	T	95

Since the data includes the total number of boardings on the line, these values are divided by the number of stops on the relevant line and the minimum and maximum number of passengers per stop is estimated. In this way, the minimum and maximum passenger numbers per stop for the departure and return directions for winter and summer were found separately on a line basis. Upper and lower bounds for the winter departure direction are shown in Table 4. These values will be used to determine the number of passengers boarding at each stop in the simulation model.

Table 4 *- Screenshot of the passenger dataset for winter and departure direction*

Winter	start - 10.00		11.00-17.00		17.00-20.00		20.00- finish	
Departure	MAX	MIN	MAX	MIN	MAX	MIN	MAX	MIN
32	6	1	5	1	3	1	2	1
33	5	1	6	1	3	1	2	1
35	8	1	7	1	5	1	2	1
80	4	2	2	1	3	1	2	1
90	6	1	11	1	7	1	2	1
93	6	1	8	1	6	1	3	1
28T	6	1	10	1	8	2	6	1
336E	5	1	7	1	3	1	2	1
33B	5	1	4	1	2	1	1	1
33E	3	1	1	1	2	1	1	1
33ES	4	1	1	1	2	1	1	1
33TE	5	1	1	1	2	1	1	1
33Y	4	1	5	1	4	1	2	1
36KE	4	1	5	1	5	1	2	1
38E	6	1	6	1	5	1	2	1
99A	6	1	9	1	7	1	4	1

2.4.2 Data Obtained Through Open APIs

To model realistic routes and estimate energy consumption with spatial accuracy, open-source geographical data through publicly available APIs were utilized. Initially, the Istanbul driveable road network in .graphml format from OpenStreetMap using tools such as OSMnx was downloaded. This file provides a detailed topological representation of the city's streets, including intersections, road lengths, and drivable connections, which serve as the backbone of our routing simulation.

For precise route generation and distance calculations, the OpenRouteService API was employed. This allowed the project to compute realistic paths between designated stops by incorporating Istanbul's urban traffic topology, turn restrictions, and road hierarchies. The output includes the exact polyline of the route, the total driving distance, and estimated duration under standard conditions. Each route segment was identified and stored with its coordinates for further use.

Additionally,Open-Elevation API to extract elevation profiles along each generated route were integrated . Since elevation changes significantly affect electric bus energy consumption (especially on uphill or downhill segments) this data was essential for our energy modeling. Elevation at regular intervals along the route polyline, enabling slope estimation and thereby refining the power requirement calculations for each route was sampled.

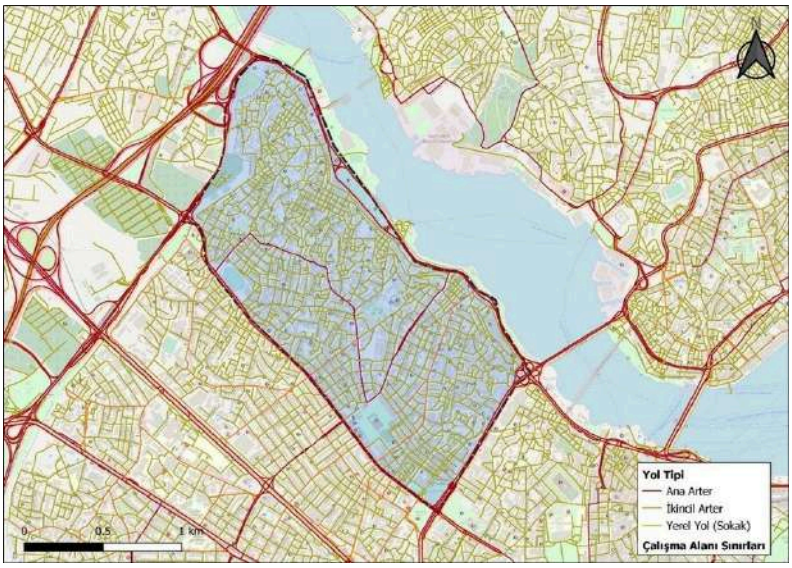


Fig.1 - *Roadmap of Fatih District*

2.4.3 Information Gathered Through External Research

In addition to the data obtained through literature research and used in the simulation, a research was conducted for the technical information of the buses to be used in the simulation. The buses that were most commonly used in the world and could be suitable for Istanbul were selected and their technical specifications were determined using the catalogs of these buses. The buses selected were electric buses BYD K8 and Karsan e-Atak, and a diesel bus Karsan Atak. These buses were used in the simulation to compare their performances in the decision-making process for the route selection.

***Table 5** - Comparison of different bus types*

	BYD K8	Karsan e-Atak	Karsan Atak
Battery Capacity (kWh)	350	220	-
CO2 Emission per kWh (kg)	0,223	0,223	3,2
Passenger Capacity	90	52	61
Bus Area	22	17	17
Photos of the buses			

In the simulation, the characteristics of the buses were derived directly from the findings of the research and implemented into the code using the information presented in Table 5.

2.5 Context Diagram and Development Workflow

The operational integration of electric buses requires a complex system architecture where multiple stakeholders exchange data. The context diagram (Figure 1) summarizes the information flow of the bus operations system with passengers, drivers, energy supply stations, maintenance team and the fleet investment division. Two of these five external entities are outside the scope of our project, which are the maintenance team and the fleet investment division, while the other entities of interest will be providing inputs.

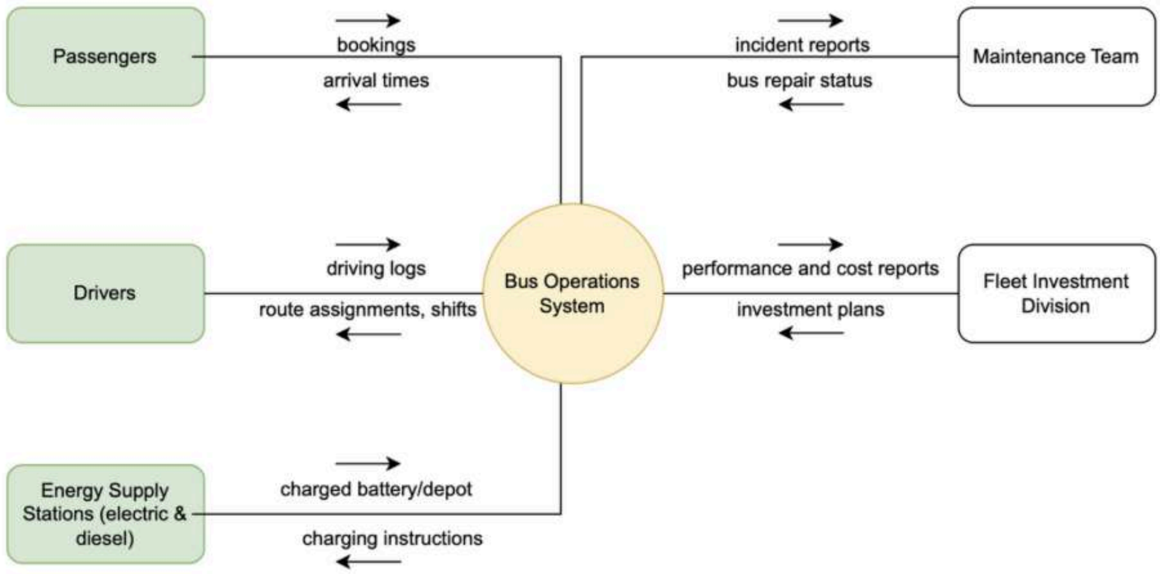


Fig.2 *- Bus Operations System Context Diagram*

The study consists of three main stages which are method & route selection, simulation and results overview. First, the operational and charging strategies of electric buses were examined by reviewing the literature of similar studies. Then, by contacting IETT, passenger density, stop coordinates and line information data were obtained, and slope profiles and route distances data were obtained separately for each route via open API.

In the second stage, a route-based simulation model was developed based on the obtained data and different scenarios were tested using the SimPy library. In this model, variables such as vehicle type, battery capacity, season, traffic and passenger density were taken into account.

In the last stage, the outputs obtained from multiple simulation iterations were analyzed to determine which lines were most suitable for conversion to electric buses. As a result of this analysis, classification was made as priority lines, second priority lines and lines that were not suitable for conversion.

2.6 Performance Criteria

To evaluate the performance of different electric bus route and fleet configurations, a set of operational and environmental metrics have been defined. These indicators guide the decision-making process by quantifying the feasibility, efficiency, and sustainability of each scenario. The selected performance criteria and their definitions are provided below:

Schedule Feasibility Rate: Indicates whether the proposed schedule was successfully completed without any route being interrupted due to battery depletion or time constraints. A feasible schedule ensures continuity in public service.

Critical Battery Threshold Compliance: Evaluates whether the battery level ever falls below the predefined safety threshold (15%) before a charging opportunity. This metric ensures operational safety and protects battery health.

Passenger Demand Fulfillment Rate: Measures the extent to which passenger demands are met throughout the route network. It is calculated as the ratio of total transported passengers to total passenger demand.

Total CO_2 Emissions: Represents the environmental footprint of the scenario. Although electric buses have zero tailpipe emissions, indirect emissions from electricity production are included using emission factors based on Turkey's grid composition.

Average Charging Frequency per Day: The average number of charging sessions needed per bus in a day. Lower values indicate better scheduling and energy efficiency, whereas higher values may imply more operational interruptions.

Fig.3 - *Decision flow for route and bus selecting*

3. Analysis for Solution/Design Methodology

3.1 Literature Overview

A literature review was conducted to establish a simulation model. It was important to find studies that contained similar examples for the parameters that were important in the simulation, and for this reason, papers covering many different areas were examined.

The International Energy Agency's report is examined to explore global examples of electric bus fleet integration and compare applications in different countries. When looking at the general situation in the world, it is seen that many countries are increasing the use of electric buses in line with their decarbonization targets in urban public transportation systems. As of 2024, there are approximately 635,000 electric buses in active service worldwide, accounting for approximately 20% of the global bus fleet. (IEA, 2024)

As shown in Figure 4, the most widespread use of electric buses globally is in China. By 2024, approximately 99% of the global e-bus fleet is in China. Particularly, the city of Shenzhen converted its entire bus fleet to electric in 2017 and now reduces CO_2 emissions by approximately 1.35 million tons annually with over 20,000 electric buses (IEA, 2024). The Shenzhen example demonstrates that large-scale integration can yield effective results in terms of environmental gains.

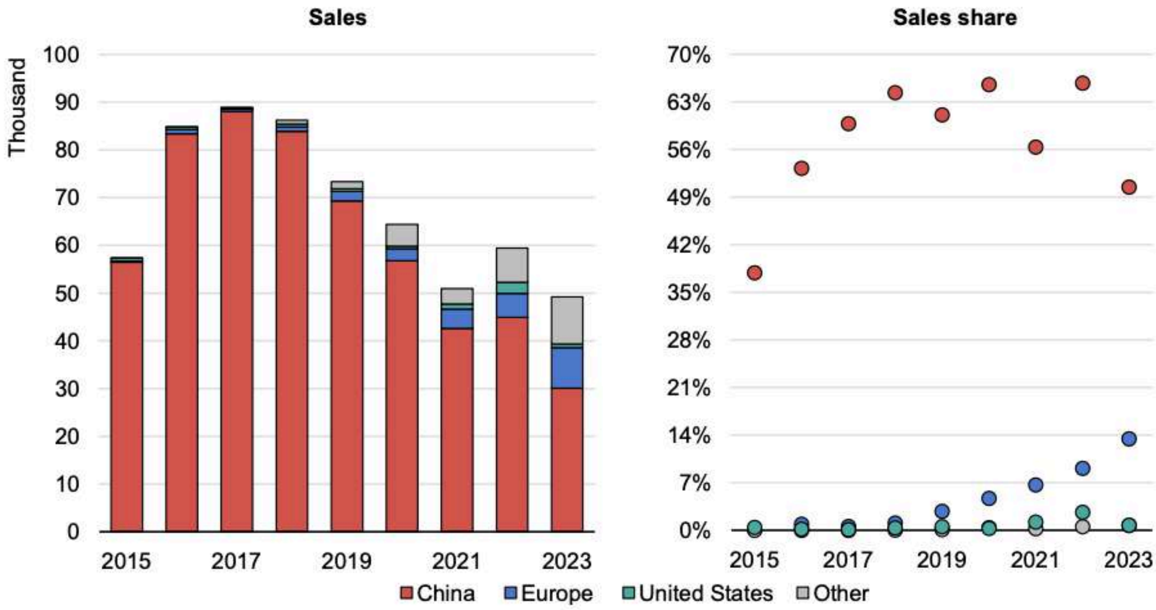


Fig.4 *- Electric bus sales and sales share by region, 2015-2023 (IEA, 2024)*

In the US, the transition is more gradual. In New York, 10% of the bus fleet is electric by 2024 and a full conversion is targeted by 2040. Similarly, in Japan, Tokyo has a 15% electric bus ratio by 2024 and has set the same target year.

In Ahmedabad (India), the battery swapping model is an alternative method implemented which is delivering charged batteries to buses by motorcycles to reduce downtime and increase operational efficiency. This application offers an alternative solution to prevent capacity losses due to charging time on busy lines. (ITDP, 2022)

Jiménez and Román (2016) suggest the fleet assignment model based on the kinematic performance of the vehicle on a specific route. Increasing emission levels and fuel consumption are significant factors for the e-bus transition process. A simulation model can be developed based on the kinematic behaviors of drivers, macroscopic variables of selected routes, and vehicle-related restrictions. To determine the driver's behavior, microcycles, meaning the time interval between two consecutive stops should be defined. To base the simulation, a sample area or depot can be selected and each route in the area can be clustered according to similar attributes such as the slope of consecutive stops, total distance traveled and average kinematic behavior during each route.

For the boarding and alighting time calculations, based on several experiments performed, Fernández (2011) reported average boarding and alighting time per passenger with respect to on-board density. The experiment also tests the effect of the different types of doors and platforms. The outcomes of this experiment shows that the door width has a greater impact than the platform height. One unexpected finding of this experiment is that boarding time increases proportional to the passenger density as shown in Fig 5. whereas alighting time increases exponentially as on-board density increases.

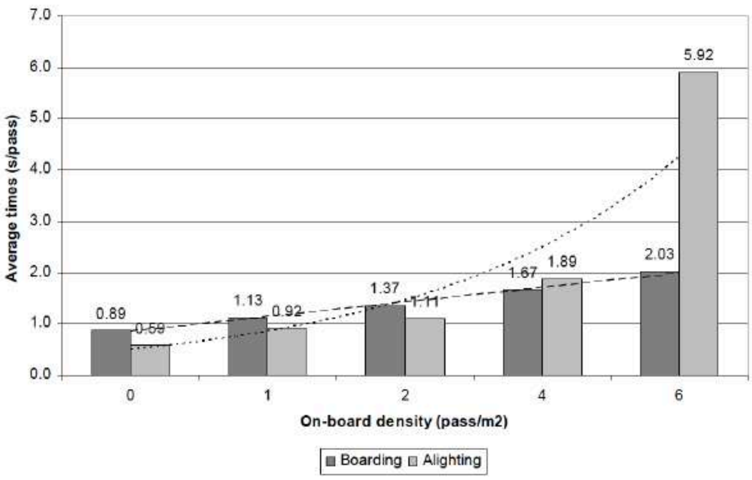


Fig.5 - *Influence of passenger density on boarding and alighting times*

Yang, Wan, and Jin (2024) New York City case study was analyzed in the literature review to evaluate the effects of switching to electric buses. In this case study, it was seen that switching to electric buses significantly reduced urban air pollution by completely eliminating exhaust emissions. According to the projection made for the year 2033 in the case study, there was a decrease of 1356 tons per year in NOx, 26.3 tons per year in PM2.5 (Particulate Matter) and a decrease of 390570 tons of CO2 equivalent per year in GHG (greenhouse gases) as shown in Table 6. (Yang, Wan, and Jin, 2024)

Table 6 - *Ecological benefits of an all-electric bus fleet in NYC (Year 2033)-projected**

Pollutant	NOx	VOC	PM2.5	PM10	SO2	GHG	Total
Emission Reduction (ton/year)	1356	13,8	26,3	42,1	6	390570	--
Economic Benefit ('000 USD/year)	32348	18,6	11206	1052	72,9	31246	75942

Furthermore, according to the same paper, electric buses have a 60-70% lower carbon footprint, including production and charging, based on a life cycle analysis, and in terms of noise pollution, electric engines operate 50% quieter than diesel engines. (Yang, Wan, and Jin, 2024)

When looking at the economic impacts, electric buses have 70% less energy costs compared to diesel buses and 30-50% lower maintenance costs thanks to simplified drivetrains. Thanks to emission reduction and operational efficiency, a long-term return of 75.9 million USD in savings per year is projected by NYC. Thanks to this study, it has been observed that the transition to electric buses is important and what its effects will be.

Significant actions have been taken by global organizations aiming to achieve environmental sustainability by reducing greenhouse gas emissions in the last decade. The Paris Agreement was negotiated in 2015 by 196 international parties under the coordination of the United Nations. This legally binding treaty requires its parties to start economic and social movements to limit the global temperature increase below 2 Celsius degrees (Matemilola, 2023). Additionally, C40, which is a global network for mayors of the most important cities, formed the Climate Leadership Group with the aim to reduce city-wide greenhouse gas emissions to zero by 2050. Istanbul Municipality is a part of C40, and also legally bound to the Paris Agreement. As stated in Istanbul Municipality's (IBB) Action Plan for Climate Change Report, IBB is planning to transform İstanbul to a carbon-neutral city by 2050.

Reducing greenhouse gas emissions city-wide is a multi-faceted task including constant energy consumption, public transportation, and waste which can be seen in Figure X (IBB Climate Change Action Plan Report, 2021). The former contributes to the major portion of emissions, and is caused by the energy consumption of factories and buildings all around the city. The transportation sector is the second major contributor to the greenhouse gas emissions city-wide. Therefore, IETT focuses on reducing CO_2 emissions for public transportation by deploying electric vehicles into its fleet.

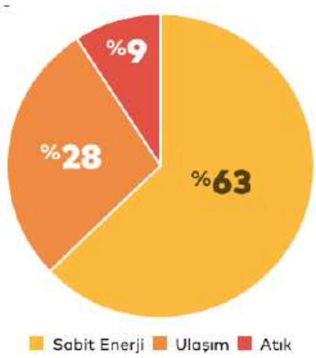


***Fig.6** - Distribution of Emission Percentages for Different Sectors in Istanbul*

Edwards, Larivé, and Beziat (2011) establish key conversion parameters for diesel bus energy usage. These parameters are applied in evaluating simulation outputs to compare electric buses with each other and the diesel buses with a suitable conversion factor. In the paper, the energy content per unit kilogram of diesel fuel is calculated based on the Lower Heating Value (LHV) and density range. In this context, the LHV value of diesel is taken as approximately 11.97 kWh/kg, and its density is in the range of 0.82-0.845 kg/L, as shown in Table 7; as a result of the conversions made, the energy content is calculated in the range of 9.8-10.1 kWh/L. In the analyses conducted within the scope of this study, the energy equivalents of diesel buses were determined by taking the mentioned conversion range as reference. (Edwards, Larivé, and Beziat, 2011)

***Table** 7 - Energy content of diesel fuel used for conversion*

	Unit	Diesel
Density	kg/L	0,82
(changes between)	kg/L	0,845
LHV	MJ/kg	43,1
	kg/kWh	0,084
	kWh/kg	11,97
Energy Calculation	kWh/L	9,8
(changes between)	kWh/L	10,1

In (Wikner & Thiringer, 2018), the aging models for lithium-ion batteries were studied. Electric vehicles use different types of batteries as their power source such as lithium-ion and lithium-iron phosphate being most common. Therefore, ensuring the long-term usage of these batteries is crucial to sustaining the quality of the existing public transportation. To achieve this, factors that accelerate battery aging should be minimized. These factors include the environment temperature, State of Charge (SOC), and charge rates. SOC is the remaining capacity level of a battery right before charging. While high temperatures and charge rates accelerate the aging of the battery, the same linear relationship cannot be used to explain the relationship between aging and SOC levels. In this project, the literature regarding the relationship between lithium-ion batteries' SOC levels and battery health was analyzed since lithium-ion batteries are the most commonly used battery type in electric buses. In (Wikner & Thiringer, 2018), it was stated that the battery aged the fastest with both high and low SOC levels. If the SOC of a lithium-ion battery were to decrease to a level that is less than 15% before charging, the life of the battery would decrease faster. Therefore, it is crucial to keep the minimum battery SOC levels around 15% during electric bus operations to ensure long battery life.

The energy consumption of electric buses is significantly influenced by road gradient, a critical factor in the planning and operation of electric public transportation systems, particularly in cities with varying topography such as Istanbul. Empirical studies consistently show that energy demand increases sharply as slope rises.

Abdelaty and Mohamed (2021) developed a predictive model for battery electric bus energy consumption, identifying road grade as one of the most influential variables. This study supports the idea that steeper roads lead to significantly higher operational demands for electric buses.

Further, Li (2023) investigated energy performance under severe gradient conditions and found that electric vehicle energy consumption could double on gradients steeper than 6°, approximately equivalent to a 12% slope. Although data for slopes between 12% and 15% is limited, this nonlinear increase strongly suggests that significant penalties occur as slope rises beyond moderate thresholds. Doubling threshold of model set to 10% slope derived from these studies

Several studies emphasize the asymmetric impact of road gradient on electric vehicle energy consumption. Uphill segments significantly increase energy demand due to gravitational resistance, while downhill segments don't reduce the consumption with the same ratio as uphill segments increase . This is largely attributed to the continuous energy use of auxiliary systems. Moreover, when maintaining a constant speed — a common assumption for realistic traffic modeling — uphill driving demands more power, whereas downhill driving still requires power to overcome friction and maintain the speed. (Ahıska, Ozgoren, & Leblebicioglu, 2021)

Figure 7 from (Ahıska, Ozgoren, & Leblebicioglu, 2021) clearly illustrates this asymmetry in energy behavior across a 1000-meter segment with approximately ±10% slope.

While the uphill scenario (left) leads to a steep increase in total energy consumption and a sharp decrease in battery state-of-charge (SOC), the corresponding downhill segment (right) does not offer a proportional reduction. For instance, if climbing a 10% slope increases consumption by 30% (e.g., from 1.0 to 1.3), descending the same slope only yields a modest reduction (e.g., from 1.0 to ~0.9 instead of 0.7). This nonlinear response highlights the importance of slope-aware route planning in electric bus operations.

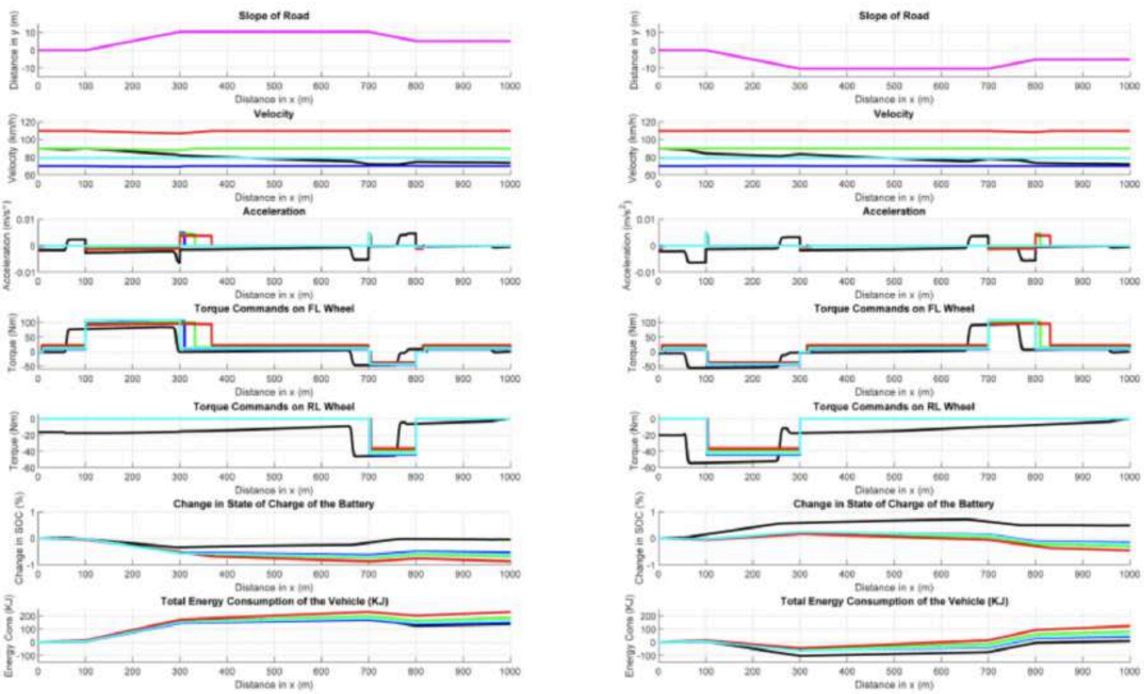


***Fig.**7 - Study on electric consumption related to slopes*

According to Doulgeris et al. (2024), absolute energy consumption per km is linked to slope and vehicle load variation. In Fig.8, the effect of vehicle mass excluding slope and air conditioning demand was indicated as the black line. Additional load in the vehicle increases the energy consumption per km slightly. This effect further increases when combined with the increasing percentage slope on the itinerary.

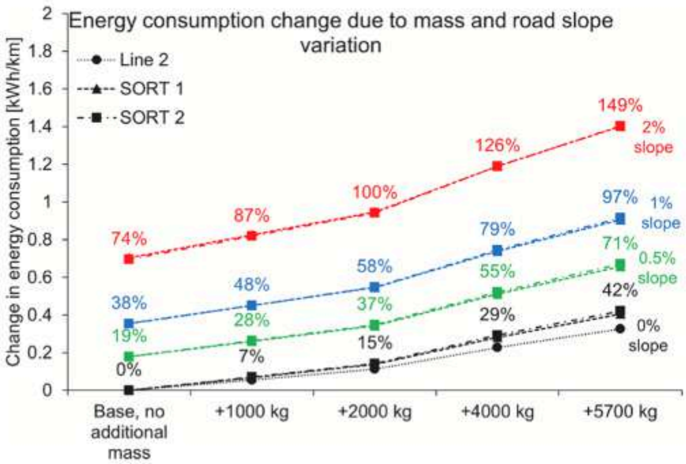


Fig.8 *- Energy consumption variation due to mass and road slope variation expressed as the absolute value (kWh/km) of the additional energy consumption. Data labels (coloured text boxes) also present the respective percentage increase of the energy consumption due to the mass and slope variations.*

In addition to vehicle-load and slope effects, another important finding of Doulgeris et al. (2024) is the strong correlation between ambient temperature and energy consumption. Validating the simulation with different bus types, itinerary distances and test days, they put forward that deviation from the ambient temperatures around 15-20 C° leads to an increasing demand for air conditioning, which in return soars energy consumption, as shown in Fig.9.

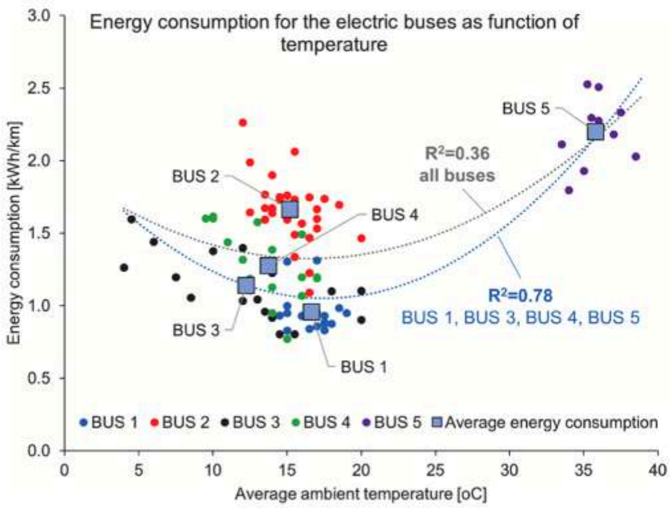


Fig.9. *- Measured daily energy consumption for all the buses, presented as a function of ambient temperature. Grey trend line was created using the measured energy consumption from all the buses, while blue trend line was created with the data from BUS1, BUS3, BUS4 and BUS5*

Similarly, Zacharof et al. (2023) also ran a simulation using the measurements from the Zincirlikuyu-Avcilar route in İstanbul to capture energy consumption changes due to the heating, ventilation, and air-conditioning (HVAC) effect during summer and winter periods. As well as temperature increase, solar radiations also affect the HVAC requirement during summer periods. On average, HVAC accounts for 11% of the total energy consumption in summer and 25% in winter. Their investigation also uncovers the link between passenger density on board and the energy consumption level. According to their findings, there occurs a 10% increase in energy consumption between 10 and 45% passenger density levels. In addition, they adopt a linear regression model for CO2 emission with temperature, solar radiation, and percentage passenger occupancy.

In the design of energy management systems for electric vehicles (EVs), particularly electric buses used in public transit, charging strategy plays a crucial role in balancing operational efficiency and battery health. A widely adopted practice is to limit the state of charge (SoC) to approximately 80% during routine fast charging. This approach is motivated by both technical constraints and practical efficiency.

Lithium-ion battery charging follows a two-phase process: constant current (CC) and constant voltage (CV). During the CC phase, charging occurs rapidly and efficiently. However, once the battery reaches around 80% SoC, it enters the CV phase, where charging power progressively decreases to prevent overheating and degradation (Lu et al., 2013). In this phase, charging the remaining 20% can take nearly as long as the initial 80%, significantly reducing turnaround efficiency.

Numerous studies and industry standards have pointed out that fast charging to 80% typically requires 15 to 30 minutes, while charging to 100% may take over an hour, particularly for large-capacity batteries used in transit buses (IEA, 2022; Xie et al., 2020).

Beyond energy inefficiency, deep and repeated full charges can accelerate battery aging, decreasing long-term reliability. Consequently, most fleet operators adopt an 80% charging limit for fast interim charging, reserving full charges for overnight depot scenarios when time constraints are minimal.

In the simulation model, this operational reality by assuming that buses, when returning to the depot for recharging during the day, only charge up to 80% SoC was incorporated. This reflects both industry best practices and physical constraints of current battery and charging infrastructure. It also ensures that modeled operations remain feasible within realistic time frames and charging station availability.

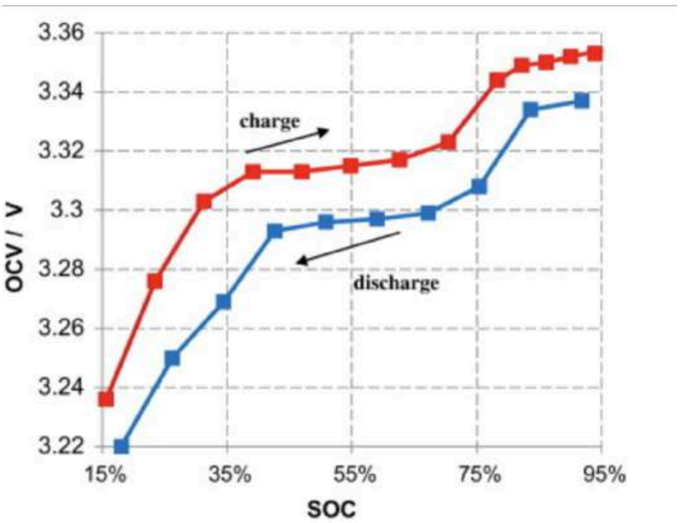


Fig.10 - *OCV–SOC relationship during charge and discharge, showing hysteresis behavior in Li-ion cells.*

3.2 Chosen Simulation Model for the Study

3.2.1 Brief Overview of the Selected Approaches

To systematically evaluate the feasibility and performance of integrating electric buses into Istanbul's public transportation network, a multi-stage methodology combining technology assessment, battery and charging strategy selection, geospatial data processing, and simulation modeling were developed. This approach ensures that the analysis captures both the operational and environmental aspects of electric bus deployment under realistic urban conditions. The methodology begins with a comprehensive technology review, which informs the selection of battery chemistries and charging strategies suitable for Istanbul's transportation infrastructure and energy landscape. Next, spatial and topographical data are integrated to accurately model route conditions, including distance and elevation, which directly impact energy consumption. These inputs are then embedded into a discrete event simulation framework, which captures the dynamic behavior of electric bus operations over time—considering factors such as charging cycles, route scheduling, and vehicle availability. This end-to-end methodological flow allows for both technical rigor and adaptability to evolving technological trends, providing a robust foundation for evaluating optimal routes and fleet configurations.

As an initial step in methodology, a comprehensive technology review was conducted to assess the current state and future direction of electric bus systems. A technology review, in the context of transportation planning and systems engineering, involves systematically evaluating existing and emerging technological components that influence the feasibility, performance, and scalability of the solution. Within the scope of the project, this review served as the foundation for modeling operational scenarios, selecting candidate vehicles for simulation, and understanding constraints related to energy infrastructure.

A central element of technology review was battery technology, as it directly affects range, energy consumption, charging strategy, and environmental impact. Electric buses primarily rely on lithium-based battery systems due to their high energy density and operational maturity. The most widely deployed battery type in commercial e-buses today is the Lithium-Ion (Li-ion) battery. These batteries offer favorable characteristics such as high energy-to-weight ratio, relatively fast charging capabilities, and acceptable cycle life. Within this category, Lithium Nickel Manganese Cobalt Oxide (NMC) and Lithium Iron Phosphate (LFP) chemistries are the most common. NMC batteries are preferred for their higher energy density, while LFP batteries offer superior thermal stability and longer lifespans, making them attractive for heavy-duty applications.

In addition to mainstream Li-ion systems, the potential of solid-state batteries (SSBs) were examined, which represent the next generation of battery technology. SSBs replace the liquid electrolyte found in conventional Li-ion cells with a solid electrolyte, resulting in higher energy density, improved safety, and reduced fire risk. Although still under pilot-scale development for large vehicles, SSBs are projected to become commercially viable within the next decade. Their integration into public transportation could significantly increase range while reducing the need for frequent recharging.

Through this technology review, battery characteristics that formed the basis for simulation parameters were established. This ensured that route evaluations reflect real-world performance metrics and are adaptable to evolving technological trends.

In parallel with battery technologies, the choice of charging strategy plays a critical role in determining the feasibility and efficiency of electric bus integration into existing public transportation systems. Charging infrastructure not only dictates route planning and depot design but also impacts turnaround times, fleet scheduling, and total cost of ownership. As part of technology review, three main charging strategies that commonly used in electric bus operations were adapted: depot charging, opportunity charging, and battery swapping.

Depot Charging is the most widely adopted strategy globally. Buses are charged overnight at central depots using slow to medium-speed AC or DC chargers. This approach benefits from operational simplicity and lower infrastructure costs, as vehicles are charged during off-peak hours when electricity rates are lower and vehicle demand is minimal.

However, depot charging requires large battery capacities to ensure that buses can operate a full day's schedule without returning for recharge, increasing vehicle cost and weight.

Opportunity Charging refers to high-power, short-duration charging events that occur during layovers at bus terminals or designated points along a route. Using DC fast chargers, buses can partially recharge within 5 to 15 minutes, reducing required battery capacity and enabling continuous operation on high-frequency routes. This strategy, however, demands significant infrastructure investment and careful route design to synchronize charging with schedule breaks.

Battery Swapping is an emerging solution wherein depleted batteries are replaced with fully charged units at automated stations. This process takes only a few minutes and virtually eliminates downtime. It is particularly attractive for high-utilization fleets and cities with limited grid capacity. Nonetheless, battery swapping systems face challenges related to standardization, capital cost, and the logistical complexity of managing spare battery inventories.

For the simulation model, interim depot charging up to 80% state of charge as opportunity and to 100% overnight were adopted as IETT required so. , reflecting current industry best practices and infrastructure limitations in Istanbul. This strategy aligns with the operational model of IETT and supports a scalable and cost-effective transition to electric bus fleets.

Fig.11- *Comparison of technologies*

Accurate spatial data is fundamental for modeling real-world bus operations, particularly in the context of energy consumption simulations where route length and slope directly impact vehicle performance. For this reason, a key component of technology review involved sourcing and processing geospatial information to accurately model route topography and distance.

Primary data source was the Istanbul Electric Tram and Tunnel Company (IETT), which provided detailed GPS coordinates for bus stops across all routes within the scope of the study. These coordinates served as the foundation for calculating elevation profiles and inter-stop distances—two critical inputs for estimating the energy required by electric buses under various load and slope conditions.

To enrich and operationalize this data, two open-source geospatial APIs were utilized: OpenElevation API and OpenRouteService API. The OpenElevation API returns elevation data (in meters above sea level) for any given set of coordinates. By querying consecutive bus stop coordinates, elevation differences were compıuted and thus derived slope percentages for each segment of a route. This allowed the project to integrate realistic gradient effects into simulation models.

The OpenRouteService API, built on OpenStreetMap data, provided accurate travel distances and routing paths between coordinates. Unlike straight-line (Euclidean) distance calculations, this API returns the actual road distance and directionality, accounting for the true path a bus would travel. This was essential for both validating route lengths and for realistically estimating energy consumption, since electric buses expend more energy over longer and sloped paths.

By integrating IETT's bus stop data with these external geospatial APIs, the project was able to construct a high-resolution, route-specific profile of each bus line. This enabled simulations to closely reflect operational realities and support informed route prioritization for electric bus deployment.

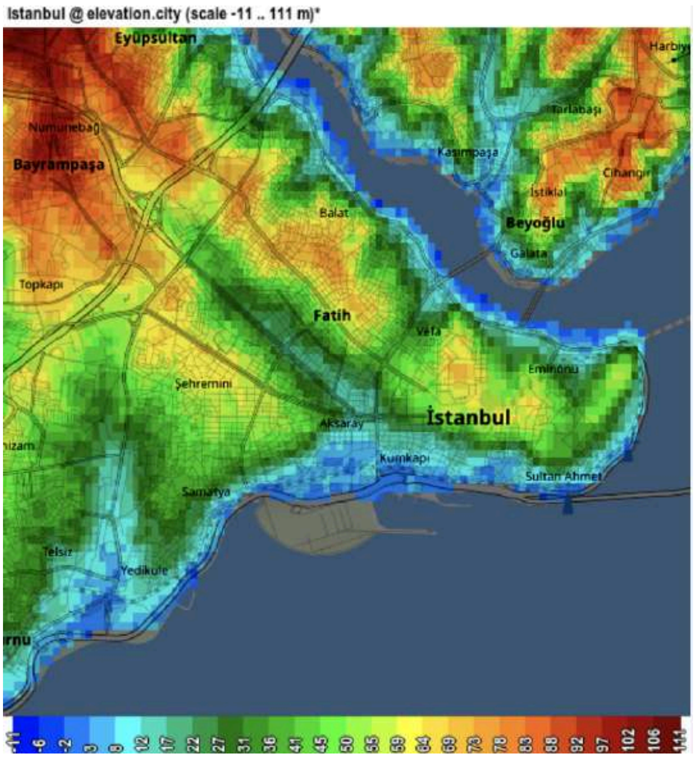


Fig.12- *Topology of Fatih District from OpenElevation*

3.2.2 Method Selection

Studies focused on the electrification of diesel-powered bus lines include a wide range of methods such as multi-criteria analysis, and simulation. An example of the former was implemented in (Emami, 2022) which evaluated scores for each bus line's feasibility for electrification regarding the criteria including avoided CO_2 emissions by electric buses and the proportion of distance covered within the pilot area. However, this method required a small subset of experts to assign weights to the decision criteria prior to route selection, making results highly dependent on the chosen subset of experts' opinions. Furthermore, this approach doesn't consider dynamic traffic variability or weather conditions. As a result, multi-criteria analysis was discarded as a potential method for this project.

In contrast, utilizing simulation is beneficial for assessing the feasibility of routes for electrification due to its ability to reflect the dynamic nature of public transportation. An electric bus' operation is affected heavily not only by routes' geographical properties but also by dynamic external factors such as traffic conditions and passenger distributions of the routes. Simulation methods can use actual data including road distances, slopes and passenger boardings, and assess how different types of electric buses would operate under varying seasonal and operational conditions. By using a simulation, multiple scenarios can be tested and performance indicators such as CO_2 emissions can be tracked.

The simulation method includes various types such as agent-based modeling, system dynamics, and discrete event simulation. System dynamics is used to explain the causality and feedback between the system components. It is not utilized for systems where operational events and delays are the area of interest. Agent-based modeling was eliminated since interactions between micro-level agents (drivers, passengers, and buses) were not the main focus of this study, and the system could be simulated without modeling agent-level decision-making behavior. Due to its suitability for reflecting the sequence of operational events, delays (e.g. recharging, traveling, and boarding), and real-time interactions between the bus operations system with external entities (e.g. passengers, traffic), a discrete event simulation was developed on the SimPy library.

3.2.3 The Logic Flow Used in the Model for Bus Operations

In the simulation, an electric bus will follow the Logic diagram depicted in Figure 13**.** Each bus will start its daily operations by leaving the depot fully charged and traveling to its first stop, Topkapı. Before each round trip, the bus will check whether its battery is sufficient to cover the whole route. If the bus cannot cover the route, it will go back to the depot for recharging. Otherwise, it will continue its operations by traveling to the next stop and onboarding & offboarding passengers. Additionally, the bus will check whether its current stop is its last stop or not. It will continue its operations in the case that the current stop is not the last stop, otherwise it will start backward routing to Topkapı.

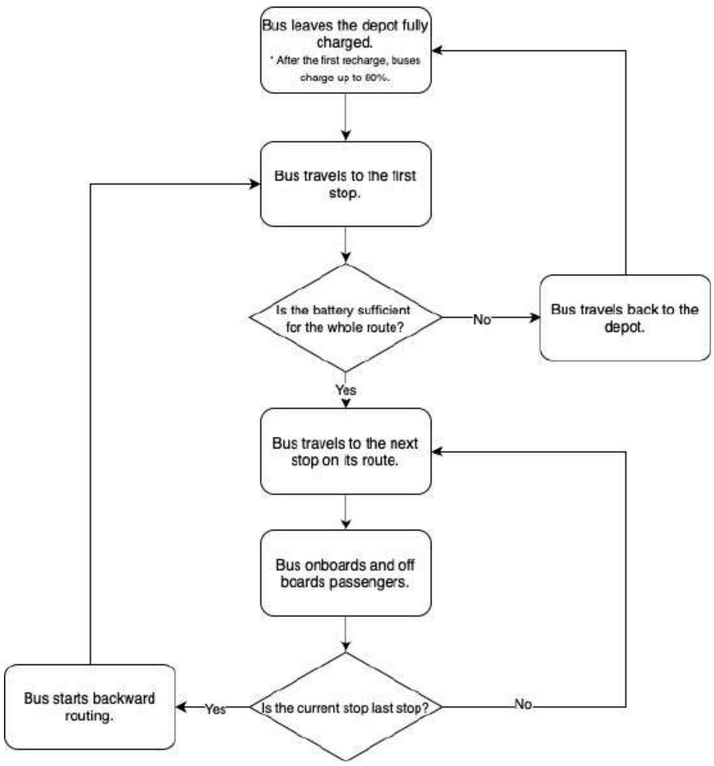


Fig.13 - *Electric Bus Logic Diagram*

A diesel bus will follow a similar logic diagram to an electric bus in the simulation as can be seen in Figure 14. However, the diesel bus will have infinite energy sources in order to prevent it from recharging at the depot since the diesel buses do not require refueling during their operation in real life.

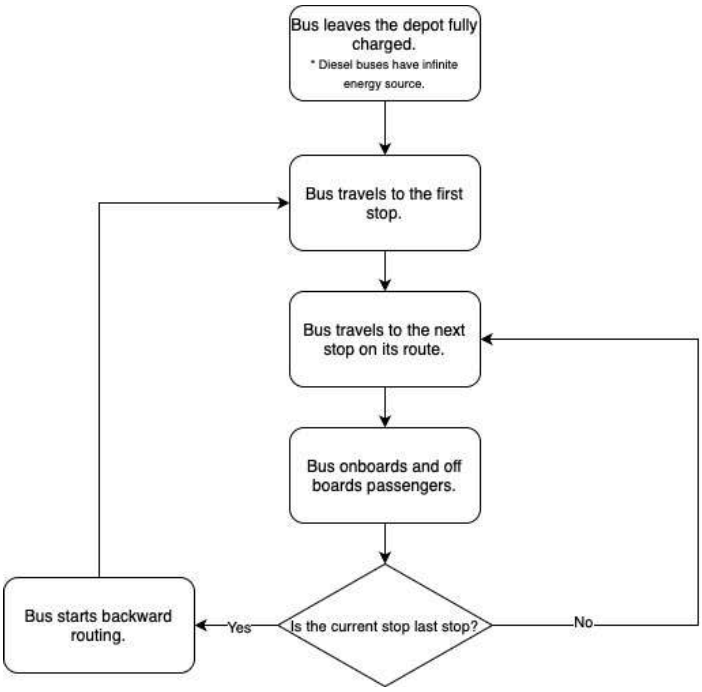


***Fig.14 -** Diesel Bus Logic Diagram*

The fuel consumption of a diesel bus is a function of travel distance, route segment slope, and passenger load. In the simulation, the base fuel consumption per passenger per km is stated as 0.04 liters according to the data obtained regarding the KARSAN ATAK's technical properties (Karsan ATAK Turkey Catalog, 2025). To account for dynamic conditions, the simulation adjusts this value by using a slope correction factor: uphill slopes will increase fuel use and downhill slopes will decrease fuel use. Then the base fuel consumption is passed to the interpolation function used in electric buses for reflecting the passenger load's effect. However, the base fuel consumption output is converted from kWh to liters by using the 9.8 kWh/L conversion factor provided in Table 7. The final consumption is calculated in liters, and CO_2 emissions of a diesel bus are calculated accordingly by multiplying the energy consumption with 2.69 kilograms of CO_2 emissions per liter (Jayaratne, 2010).

3.3 Methodology and Assumptions of the Model

After conducting a comprehensive technology review and selecting the appropriate simulation method, the simulation model was developed based on the methods and assumptions outlined below:

One of the most critical factors influencing energy consumption is the interaction between vehicle mass and road slope variation—an aspect that was overlooked in the simulation model proposed by Jiménez and Román (2016). Therefore, the linear interpolation method proposed by Doulgeris et al. (2024) was integrated into the simulation to estimate energy consumption across different slope and mass combinations. For simplicity, each passenger was assumed to weigh an average of 70 kg, as individual mass was taken into account.

Given the coordinates of each station, the slope between consecutive stations was calculated using OpenAPI. In order to integrate the effect of slope on energy consumption into our simulation, a slope correction function was defined. By introducing energy consumption correction equations for uphill and downhill, a more realistic energy consumption logic was constructed. Based on the deductions of Doulgeris et al. (2024) and Ahiska et al.(2021), even a slight change in the slope affects energy consumption dramatically when combined with passenger load. Accounting for them, a slope-based correction factor was implemented into the simulation.

In the case of a 0% slope and no passengers on board, the base energy consumption rate was assumed to be 0.8kwh/km (Rebouças & Delgado, 2023). The effect of slope over energy consumption is not the same for uphill and downhill trips (Ahiska, Ozgoren, &

Leblebicioglu, 2021). To maintain travel speeds of approximately 15, 30, and 45 km/h—depending on the time block defined in the simulation—additional energy is required during uphill travel, whereas a non-negligible amount of energy is still consumed when traveling downhill. Therefore, adjustments to the slope must be made based on the slope levels for both uphill and downhill segments. To account for the impact of slope and passenger load on energy consumption—while also considering both uphill and downhill trips—a slope correction was initially applied as described below. When the slope is positive, meaning the e-bus climbs uphill, the slope correction parameter is derived as follows (Ahiska, Ozgoren, & Leblebicioglu, 2021) :

$$\text{Slope correction for uphill trips} = \left(1 + \frac{(\text{current slope in }\%)}{10}\right)$$

Eq.1 *- Equation of the slope correction parameter for uphill trips*

When the slope is negative, meaning the e-bus travels downhill, the slope correction parameter is derived as follows (Ahiska, Ozgoren, & Leblebicioglu, 2021) :

$$\text{Slope correction for downhill trips} = max\left(0.7\ , 1 + \frac{(\text{current slope in }\%)}{25}\right)$$

Eq.2 *- Equation of the slope correction parameter for downhill trips*

As given in Eq. 2, during downhill trips, the corrected slope does not fall below 70% of the actual slope. This constraint was imposed to ensure that the energy consumption remains at a certain level, even when the bus was traveling downhill.

After the slope correction step, the effect of passenger load and slope over energy consumption rate was calculated. To do so, the slope- and passenger load-dependent variation in energy consumption presented in Fig.8 was utilized. The passenger load on board was calculated dynamically at each stop. Based on the passenger load between two consecutive stops, a linear interpolation was performed to estimate the energy consumption rate for a given passenger load.

$$b(m) = \begin{cases} 0.8 + \frac{0.9-0.8}{1000-0} \cdot (m-0) & \text{if } 0 \leq m < 1000 \\ 0.9 + \frac{1.0-0.9}{2000-1000} \cdot (m-1000) & \text{if } 1000 \leq m < 2000 \\ 1.0 + \frac{1.2-1.0}{4000-2000} \cdot (m-2000) & \text{if } 2000 \leq m < 4000 \\ 1.2 + \frac{1.4-1.2}{5700-4000} \cdot (m-4000) & \text{if } 4000 \leq m \leq 5700 \\ 1.4 & \text{if } m > 5700 \end{cases}$$

Eq.3 - *Base energy consumption rate equations based on different passenger load levels*

In Eq.3, *b(m)* indicates the base energy consumption rate given the passenger load and *m* is the passenger load in kg. The final energy consumption level was calculated by multiplying the corrected slope and *b(m)*.By following these steps, the simulation dynamically adjusts the energy consumption level across each consecutive stop.

Passenger distribution is one of the most critical factors in simulation results and in determining line-based decision-making mechanisms. Since the data-set provided by IETT only includes the total number of passengers per-trip, stop-level passenger numbers were estimated using a uniform distribution. For each selected line, the day was divided into time blocks, and the minimum and maximum total number of passengers transported during each time block was calculated. These values were then divided by the number of stops on the respective line to obtain the minimum and maximum number of passengers that could board at each stop. The number of passengers boarding at each stop was then randomly selected from within this minimum-maximum range.In addition, these passenger distributions were calculated separately for the summer and winter seasons so that seasonal demand variations in passenger patterns could be integrated into the simulation. The number of alighting passengers was determined by this uniform assumption. The number of boarding passengers, on the other hand, was determined by random assignment within the identified range, limited by the number of passengers on board.

To incorporate the variations in traffic conditions throughout the day, the day was divided into four time periods: morning, afternoon, evening, and night. As previously mentioned, this allowed the model to account for changes in passenger distribution during peak travel periods with higher demand.

Another variable influenced by the time of day is traffic. Based on the typical behavior of traffic flow, it was assumed that average travel speed decreases during periods of heavy traffic and increases during periods with lighter traffic. Therefore, the average travel speed is changing around 15 km/h during the morning and evening time blocks, 30 km/h during afternoon, and 45 km/h during the night time block, reflecting typical traffic conditions for each period. Varying average speeds added additional randomness to the simulation.

According to the information provided by IETT, diesel vehicles are maintained and fueled in garages in certain regions, therefore, charging stations for e-buses will be installed in garages. It was suggested that Topkapı garage should be prioritized for use in the simulation, and all buses were taken to Topkapı garage for charging in the simulation. However, no limit was set for the number of buses going to the garage at the same time. An assumption was made that the garage area and the number of charging stations are unlimited. Since the number of charging stations also requires an investment cost, it falls outside the scope of the project. For this reason, a plan was established in which the bus going to the garage starts charging directly without waiting in line. Therefore, considering the nature of this study and the initial phase of the transition to electric buses, it was assumed in the simulation that buses would be charged exclusively at the depot, based on the premise that depot-charging will be used during the early stages of implementation.

Each bus begins its operation with a fully charged battery (100% state of charge). After the completion of each trip, the remaining battery level is checked to determine whether it is sufficient to complete the subsequent trip. The remaining battery level check was made based on the average energy consumption values specific to the relevant route. If the current battery level is insufficient for the next trip, the bus is routed directly to the depot for recharging.

At the depot, it was assumed that the battery would be charged to 80% capacity within one hour. Waiting for the battery to reach 100% state of charge was considered inefficient; thus, resuming operations after reaching 80% was deemed sufficient from a time-efficiency perspective. (Cole ,2024)(Lu et al., 2013)

In addition to this constraint, the battery level was also monitored to ensure that it did not fall below 15%, as operating below this threshold can negatively impact the long-term battery lifetime. (Wikner & Thiringer, 2018)

Boarding and alighting times were also calculated dynamically. If there were no passengers at a given stop, the bus continued without halting. In cases where passengers were present, dwell time was assumed as 8.8 seconds on average. The dwell time accounts for door opening and closing durations, as well as deceleration and acceleration times upon arrival and departure from the stop. (Fernández, 2011)

Alighting and boarding times were dynamically determined based on the findings of Fernández (2011). In scenarios with high on-board passenger density—i.e., when the bus is crowded—both boarding and alighting durations increase proportionally with the density level. Therefore, at each stop, on-board passenger density was computed. On-board passenger density was calculated by dividing the number of passengers on board by the area of the selected bus. Based on this density level, boarding and alighting times per passenger were calculated using linear interpolation as follows:

$$\text{Density } d : [0,\ 1,\ 2,\ 4,\ 6]$$
$$\text{Alighting time per passenger [sec]} : [0.59,\ 0.92,\ 1.11,\ 1.89,\ 5.92]$$

$$\text{Density } d : [0,\ 1,\ 2,\ 4,\ 6]$$
$$\text{Boarding time per passenger [sec]} : [0.89,\ 1.13,\ 1.37,\ 1.67,\ 2.03]$$

$$\text{boarding time} = N_b \cdot f_b(d)$$
$$\text{alighting time} = N_a \cdot f_a(d)$$

Where:

- N_b: number of boarding passengers at the stop, determined by the stop-level passenger distribution
- N_a: number of alighting passengers at the stop, also determined by the stop-level passenger distribution
- $f_b(d)$: interpolated boarding time per passenger as a function of on-board density d
- $f_a(d)$: interpolated alighting time per passenger as a function of on-board density d

Eq.4 - *Equations for boarding and alighting time calculations*

$f_a(d)$ and $f_b(d)$ were calculated as follows:

$$f_a(d) = \begin{cases} 0.59 + 0.33(d-0), & 0 \le d < 1 \\ 0.92 + 0.19(d-1), & 1 \le d < 2 \\ 1.11 + 0.39(d-2), & 2 \le d < 4 \\ 1.89 + 2.015(d-4), & 4 \le d \le 6 \\ 5.92, & d > 6 \end{cases}$$

Eq.5 - *Alighting time equations at each stop based on different on-board passenger density*

$$f_b(d) = \begin{cases} 0.89 + 0.24(d-0), & 0 \leq d < 1 \\ 1.13 + 0.24(d-1), & 1 \leq d < 2 \\ 1.37 + 0.15(d-2), & 2 \leq d < 4 \\ 1.67 + 0.18(d-4), & 4 \leq d \leq 6 \\ 2.03, & d > 6 \end{cases}$$

Eq.6 - *Boarding time equations at each stop based on different on-board passenger density*

Passenger boarding and alighting times were calculated accordingly and the total waiting time at each stop was determined based on these dynamic calculations as follows:

$$Total\ waiting\ time\ = \begin{cases} 0\ ,\ if\ N_a = 0\ and\ N_b = 0 \\ alighting\ time + boarding\ time + 8.8\ ,\ otherwise \end{cases}$$

Eq.7 - *Total waiting time calculation at each stop*

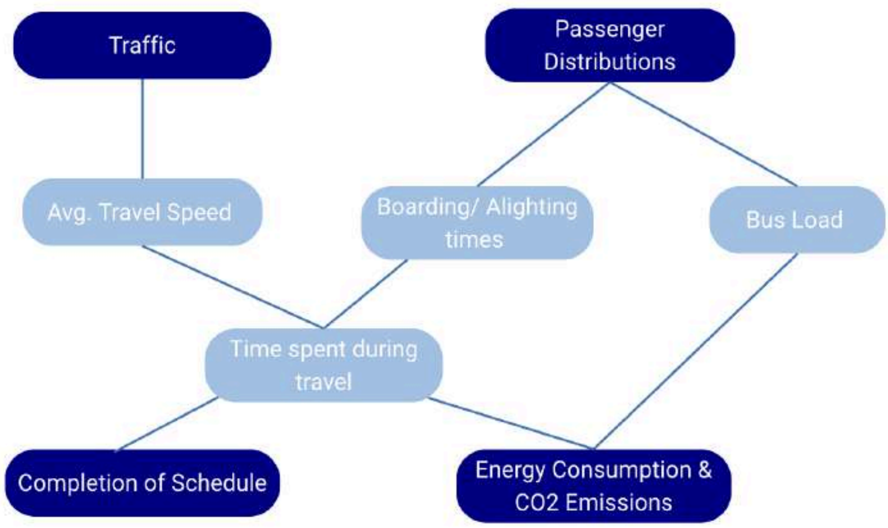


Fig.15 - *Randomness diagram of the simulation*

As demonstrated in Fig.15, the randomness in the simulation mainly stems from the uncertainty in traffic conditions and the uncertainty in passenger distribution at the stops. Other parameters affected by these factors include the bus travel speed, boarding/alighting times, bus load, and time spent during travel. The time spent during travel, in turn, affects the completion of the schedule as well as the energy consumption and CO_2 emissions.

Although electric buses operate with zero tailpipe emissions, they are not entirely emission-free when evaluated from a full energy lifecycle perspective. The carbon footprint of electric bus operations arises indirectly from the electricity used to charge their batteries. Therefore, the emissions associated with electric buses must be calculated based on the carbon intensity of the electrical grid that supplies power to the public transportation infrastructure.

In this study, an emission factor of 0.223 kg CO_2 per kWh was applied to estimate the indirect emissions produced by electric buses, being aware of the fact that IETT also wants to use renewable energy sources to power these buses. This value corresponds to the average grid emission intensity in Istanbul and represents the amount of carbon dioxide emitted for each kilowatt-hour of electricity generated. The logic behind this approach is grounded in internationally recognized carbon accounting principles, which treat electricity consumption as an emission source if the power originates from fossil fuel-based generation. Consequently, for every unit of energy consumed by the bus, the corresponding CO_2 emissions can be traced back to the national energy mix powering the grid.

This method of CO_2 estimation is supported by the International Energy Agency and academic literature, where the carbon impact of electric vehicles is routinely assessed using national or regional grid emission factors (IEA, 2022). Unlike diesel buses, which emit CO_2 directly through combustion, electric buses shift the emission source to the grid level. As a result, any decarbonization in national electricity production directly improves the environmental performance of electric bus fleets.

By integrating this emission factor into the simulation, it was ensured that energy consumption values (in kWh) could be directly translated into realistic CO_2 emission estimates for each route. This allowed for fair comparisons with diesel alternatives and enabled the project to prioritize routes that maximize emission savings through electrification.

No modifications were made to the existing schedule; the schedule provided by IETT was used as a deterministic input, and all calculations were based on a scenario in which the entire bus fleet was converted to electric vehicles. Using the total number of trips operated on the selected bus lines, the average number of trips a single bus is required to complete per day was calculated. The model then assessed whether these daily trips could be completed using an electric vehicle, and evaluated key outcomes such as total electricity consumption, CO_2 emissions, and the number of times the vehicle returned to the depot for charging. Based on these outputs, decisions were made regarding which lines are most suitable for electric bus deployment. Furthermore, an operational constraint was imposed whereby each bus—along with the same assigned driver—was assumed to be available for a maximum operational window of 18 hours per day, from 06:00 to 00:00. This period includes all activities, such as scheduled trips and return journeys to the depot for charging. Simulation results were evaluated in light of this constraint, and any bus lines requiring more than 18 hours to complete their assigned schedules were excluded from the electric bus transition scenario, as they were deemed infeasible under current operational limits.

Driver behavior was assumed to be constant for each trip, and its impact on energy consumption was considered negligible.

To account for the effect of air conditioning, the average temperatures during summer and winter months over the past 10 years was calculated. Based on the findings of Doulgeris et al. (2024) and Zacharof et al. (2023), the variation in energy consumption with respect to ambient temperature was determined. Referring to the convex trend illustrated in Fig.9, the increase in energy consumption was estimated to be 13.5% in summer and 27% in winter, based on the average seasonal temperatures in Istanbul. These multipliers were also used in the calculation of energy consumption rate for summer and winter periods.

Lastly, as an evaluation procedure, a total of 100 simulation runs were conducted. The same random seed was used for all three selected bus types in each run. Using the same random seed ensured consistent passenger distribution at stops and all other stochastic factors across the three bus types. This consistency facilitated the evaluation of differences between the bus types and allowed for a more accurate comparison of their performances.

3.4 Industrial Engineering Skills and Techniques Used to Implement the Model

The electric bus simulation model developed in this study was designed based on numerous methods and technical knowledge acquired in the Industrial Engineering discipline. Mathematical modeling, decision support systems, data analysis, simulation modeling, optimization techniques and statistical analysis methods were used in the design and implementation process of the model.

Firstly, the route-based simulation model was developed as a discrete-event simulation using the SimPy library. This approach enabled sequential and time-dependent modeling of operational events such as bus movements, charging processes, passenger boarding/alighting and garage returns. Variable parameters such as route lengths, slope profiles and passenger density were integrated into the model by taking them from different data sources.

Statistical methods were used in the model verification and validation steps; the reliability of the simulation results was tested by calculating sample means and 95%

confidence intervals on the outputs obtained in different scenarios. In addition, a two-sample t-test was applied to test the statistical significance of the differences in energy consumption and CO_2 emissions between diesel and electric buses.

As a result, the methods and tools used in this project reflect the holistic application of optimization, system modeling, statistical analysis and data processing skills acquired during the Industrial Engineering undergraduate education.

4. Development of Alternative Solutions

The development of robust alternative configurations requires formulating the electric bus deployment problem as a multi-objective decision system aligned with operational feasibility, energy efficiency, service quality, and environmental impact. This process is structured around a five-stage evaluation pipeline , where only the configurations satisfying all preceding conditions proceed to the next. Let $\mathbf{x} = [x_1, x_2, \ldots, x_n] \in \mathcal{X}$ denote a feasible configuration vector, where each x_i encodes the bus schedule, number of allocated vehicles, intermediate charging points, and SOC thresholds on route i. , define a vector-valued objective function:

$$\min_{\mathbf{x}\in\mathcal{X}} \mathbf{F}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ f_2(\mathbf{x}) \\ f_3(\mathbf{x}) \\ f_4(\mathbf{x}) \\ f_5(\mathbf{x}) \end{bmatrix}$$

where each $fj(x)$represents one of the five decision criteria:

1. Schedule Feasibility Formulation:

$$f_1(\mathbf{x}) = \sum_{i=1}^{n} [1 - \phi_i(x_i)]$$

- $\phi_i(x_i) \in \{0, 1\}$: Feasibility indicator (1 if route i is completed successfully)
- Elimination rule: $f_1(\mathbf{x}) > 0 \Rightarrow \mathbf{x} \notin \mathcal{X}_{valid}$

2. Minimum Battery Level Check Formulation:

$$f_2(\mathbf{x}) = \sum_{i=1}^{n} \max\left(0,\ 0.15 - \min_{t} SOC_i(t)\right)$$

- Penalizes configurations where SOC drops below 15% at any time t on route i
- $SOC_i(t) \in [0,1]$: Normalized battery state of charge profile
- Constraint: $\min_t SOC_i(t) \geq 0.15 \quad \forall i$

3. Passenger Demand Coverage Formulation:

$$f_3(\mathbf{x}) = -\sum_{i=1}^{n} \eta_i(x_i)$$

- $\eta_i(x_i) = \frac{\text{served passengers}}{\text{total demand}} \in [0,1]$
- Negative sign indicates that higher demand satisfaction is better (we minimize $-f_3$)

4. Total CO_2 Emissions Formulation:

$$f_4(\mathbf{x}) = \sum_{i=1}^{n} E_i(x_i) \cdot \gamma$$

- $E_i(x_i)$: Energy consumed on route i in kWh
- $\gamma = 0.223\,\mathrm{kgCO_2/kWh}$: Istanbul's estimated regional grid emission factor

5. Average Daily Charging Events

$$f_5(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^{n} C_i(x_i)$$

- $C_i(x_i)$: Number of charging sessions required per day on route i

Where

x_i denotes the decision vector for route i, which includes the schedule, the number of buses assigned, and the charging plan.

ϕ_i is a binary feasibility indicator that reflects whether the schedule for route i can be completed feasibly.

$SOC_i(t)$ represents the state of charge at time t for buses operating on route i.

η_i is the passenger demand satisfaction ratio on route i, indicating the proportion of passenger demand that is met.

E_i denotes the energy consumption on route i, expressed in kilowatt-hours $[kWh]$.

γ refers to the grid emission factor, given in units of kilograms of CO_2 per kilowatt-hour $[kg\ CO_2/kWh]$.

C_i represents the average number of charging events per day for route i.

$\mathcal{X}$ defines the feasible decision space for the optimization model.

Filtering and Dominance Strategy

Alternatives are filtered stage-by-stage as per the decision flow. Configurations that violate hard constraints (e.g., schedule feasibility, minimum SOC) are eliminated. The remaining configurations are compared using Pareto dominance:

$$\mathbf{x}^{(1)} \prec \mathbf{x}^{(2)} \iff \forall j,\ f_j(\mathbf{x}^{(1)}) \leq f_j(\mathbf{x}^{(2)}) \wedge \exists j,\ f_j(\mathbf{x}^{(1)}) < f_j(\mathbf{x}^{(2)})$$

This results in a Pareto frontier of valid configurations balancing energy, operational, and environmental goals.

Beyond the primary objectives incorporated into the current decision framework, the solution space can be further enriched by incorporating additional domain-specific criteria into the optimization formulation. This allows for the generation of higher-dimensional Pareto fronts, where alternative solutions reflect a broader range of strategic trade-offs. For instance, objectives such as battery degradation cost, charger utilization efficiency, driver scheduling compatibility, or load balancing across depots can be formalized and integrated as independent optimization goals.

Formally, let $\mathbf{F}(\mathbf{x}) = [f_1(\mathbf{x}), f_2(\mathbf{x}), \ldots, f_k(\mathbf{x})$ be a vector of k objective functions, each quantifying a unique system performance indicator. As k increases, the set of non-dominated solutions $\mathcal{P} \subset \mathcal{X}$ expands in dimensionality, offering decision-makers a more granular control over operational priorities.

Define an example objective function that captures total lifecycle-aware cost per route configuration, accounting for not only immediate energy use but also long-term degradation, infrastructure dependency, and opportunity cost.

Define:

$$f_{\mathrm{LCOC}}(\mathbf{x}) = \sum_{i=1}^{n} \left[\alpha \cdot E_i(x_i) \cdot \gamma + \beta \cdot D_i(x_i) + \delta \cdot \frac{C_i(x_i)}{U(x_i)} + \zeta \cdot M_i(x_i) \right]$$

Where

$E_i(x_i)$ denotes the total energy consumed on route i, measured in kilowatt-hours $[kWh]$.

γ is the grid emission factor, expressed in $[kg\ CO_2/kWh]$, for example, 0.223 for Istanbul.

$D_i(x_i)$ represents the battery degradation cost estimate on route i, given in euros per kilometer $[€/km]$, and is based on charge depth cycles.

$C_i(x_i)$ refers to the number of charging events per day on route i.

$U(x_i)$ is the average charger utilization rate on the charging infrastructure of the respective route.

$M_i(x_i)$ indicates the estimated maintenance cost per kilometer on route i, including factors such as driver overtime and vehicle wear and tear $[€/km]$.

$\alpha, \beta, \delta, \zeta$ are calibration weights used to reflect organizational priorities within the optimization framework.

This objective function transforms electric bus routing optimization from a narrow operational task into a full-fledged strategic lifecycle cost minimization problem. It is designed to support multi-stakeholder trade-off navigation, enabling transportation authorities to internalize externalities such as carbon emissions and battery fatigue into daily operational planning. In essence, it quantifies not only "what costs less today" but "what sustains better tomorrow."

5. Comparison of Alternatives and Recommendation

5.1 Numerical Studies or Evaluation Procedure

5.1.1 Statistical Significance of the Results

It is crucial to ensure the statistical reliability of the simulation results regarding the performance of different electric bus types on various routes and seasonal conditions to determine whether the observed differences in the simulation results were caused by interactions within the system or due to randomness.

The simulation results obtained from this study are subject to randomness caused by variables such as passenger distributions and traffic conditions across bus stops. The simulation was replicated 100 times using different seeds for each run to ensure statistical reliability. After data collection from the replications, sample mean and variance were calculated for each performance metric. Then, 95% confidence intervals were built on these metrics' sample means to analyze the existence of statistical differences between different bus types across all routes. As an example, average CO_2 emissions results with 95% confidence intervals across all the routes for different bus types are provided in Table 8.

Table 8- *CO_2 emissions results for electric and diesel bus models*

	BYD K8			Karsan e-ATAK			Karsan ATAK		
Route Code	**Mean (kg)**	**95% Confidence Interval Lower**	**95% Confidence Interval Upper**	**Mean (kg)**	**95% Confidence Interval Lower**	**95% Confidence Interval Upper**	**Mean (kg)**	**95% Confidence Interval Lower**	**95% Confidence Interval Upper**
33	106,129568	106,082055	106,177081	109,016249	108,972927	109,059571	504,649261	500,384359	508,914163
35	32,1818041	32,1176607	32,2459475	32,1804499	32,1171547	32,2437451	233,767066	227,510024	240,024108
80	39,6726291	39,6572824	39,6879758	40,7162776	40,7005944	40,7319609	155,241923	154,004221	156,479624
90	24,3265244	24,2720682	24,3809806	24,3112416	24,2614412	24,3610419	118,065995	116,280215	119,851774
93	86,8473793	86,6778492	87,0169094	89,3068745	89,1683024	89,4454466	589,602097	578,560745	600,643449
336E	109,498053	109,419975	109,576132	112,127911	112,059732	112,196089	568,804283	562,77076	574,837806
33B	109,042398	109,011766	109,073031	111,920426	111,891409	111,949442	441,902322	439,274189	444,530455
33E	62,5709	62,5482312	62,5935688	64,714862	64,692032	64,737692	193,723541	192,223782	195,2233
33ES	67,3165591	67,2993834	67,3337348	68,3352563	68,318596	68,3519167	227,076733	225,632827	228,520638
33TE	76,331964	76,3077396	76,3561883	79,4492914	79,4216898	79,476893	269,624993	267,276677	271,973308
33Y	90,3859626	90,3455998	90,4263255	93,2457886	93,2091466	93,2824307	405,932282	402,517225	409,34734
36KE	78,5579992	78,5269191	78,5890792	81,4537092	81,4243709	81,4830476	355,904005	353,351859	358,456151
38E	72,9559808	72,9158471	72,9961146	75,6013832	75,5633948	75,6393715	397,584928	394,102145	401,067711
99A	34,6889821	34,6035709	34,7743932	34,6732577	34,5945328	34,7519827	174,787743	172,646811	176,928674

5.1.2. Model Verification: Ensuring Internal Consistency and Correct Logic

Verification focuses on answering the question: *Is the model built right?* This stage involved systematically checking the model's implementation for correctness in logic, units, and parameter integration. Key components of verification included:

Logical Flow and Code Auditing: Every functional block within the simulation—such as passenger boarding logic, slope-based energy calculation, and charging triggers—was inspected using step-by-step outputs and in-code logging. The simulation correctly modeled bus dwell time at each stop based on real passenger load, using dynamic time adjustments derived from literature-based interpolation.

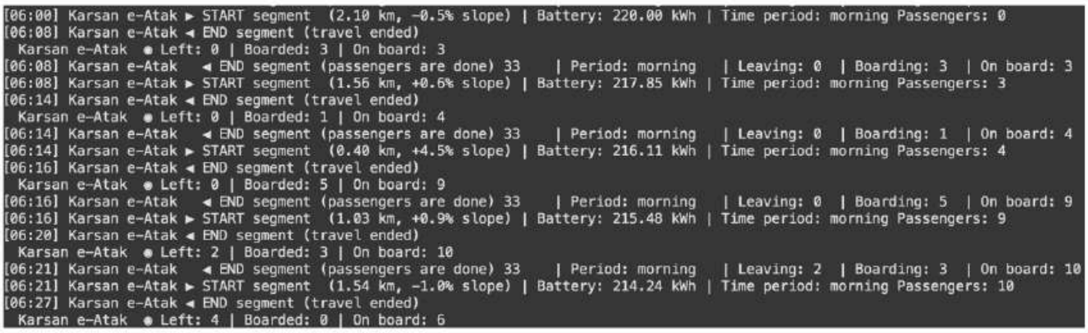


Fig.16 - *Step by step monitoring*

Energy and Charging Logic Verification: The energy consumption engine within the simulation uses linear interpolation over two dimensions (slope and passenger load) to reflect real-time energy demand. Energy depletion is updated continuously, and once a bus reaches a charging threshold, the simulation redirects it to the depot for fast charging up to 80%, in line with real-world operational behavior. This flow was verified over hundreds of replications and extreme test cases (e.g., no slope, zero passengers).

Schedule Alignment: Real-world bus schedules obtained from IETT were applied directly to the simulation. Bus routes were only included in the candidate set if they could feasibly complete trips between 06:00–00:00 with fixed vehicle count and no additional fleet support. This constraint ensured that all simulated operations reflect realistic planning constraints and service continuity obligations.

5.1.3 Model Validation: Alignment with Real-World Data and Assumptions

Validation asks: *Is the right model built? (*Barlas (1996)) that is, whether the simulation correctly replicates key dynamics of the real-world system under study. The validation process involved three levels: structural, behavioral, and policy validation.

Structural Validation Using 28T Line: The foundation of simulation validation strategy was the 28T line (Topkapı–Beşiktaş), which is an existing diesel-to-electric conversion route already in operation. All model components (stop coordinates, elevation profiles, and inter-stop distances) were directly constructed using real data for 28T. APIs such as **OpenRouteService** and **OpenElevation** were employed to extract road distances and elevation changes, enabling high-fidelity slope modeling. Battery parameters for the actual Karsan e-ATAK vehicle used on the 28T were taken from manufacturer specifications and embedded into the simulation. The route's operating characteristics served as a real-world reference point.

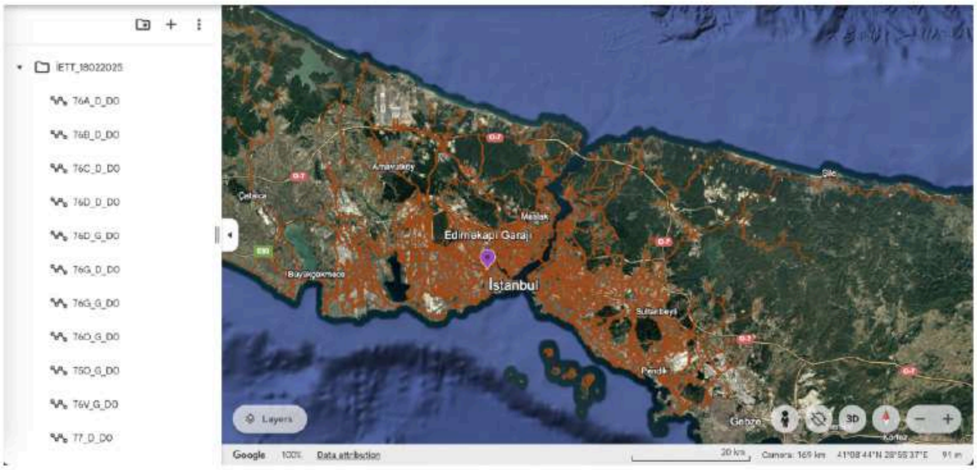


Fig.17- Screenshot from istanbul_drive.graphml

Behavioral Validation via Expected Output Trends: The simulation was observed to produce realistic, consistent behavior under varying scenarios. For instance, winter scenarios with increased HVAC use resulted in approximately 11–27% higher energy consumption, consistent with literature and manufacturer performance benchmarks. As passenger loads increased, the model correctly reflected longer boarding/alighting times and higher energy usage. Total daily energy use and battery depletion followed expected nonlinear patterns on steeper routes.

Policy/Extreme Case Testing: In line with the concept of policy validity introduced in Barlas (1996), simulation was tested in scenarios with zero slopes and zero passengers. As expected, energy usage dropped significantly, and charging behavior ceased entirely, demonstrating that the simulation behaves logically even in edge cases and can be extended to different geographies or vehicle types.

BYD K8 Simulation Summary

	Mean	Std Dev	Variance	95% CI Lower	95% CI Upper
BYD K8_Passengers	0.000000	0.000000e+00	0.000000e+00	0.000000	0.000000
BYD K8_Energy_Consumed	193.394947	5.712979e-14	3.263812e-27	193.394947	193.394947
BYD K8_CO2	45.061023	0.000000e+00	0.000000e+00	45.061023	45.061023
BYD K8_TotalServiceTime	482.592000	1.142596e-12	1.305525e-24	482.592000	482.592000

Karsan e-Atak Simulation Summary

	Mean	Std Dev	Variance	95% CI Lower	95% CI Upper
Karsan_Passengers	0.000000	0.000000e+00	0.000000e+00	0.000000	0.000000
Karsan_Energy_Consumed	196.754947	2.285191e-13	5.222100e-26	196.754947	196.754947
Karsan_CO2	45.843903	4.284734e-14	1.835894e-27	45.843903	45.843903
Karsan_TotalServiceTime	550.992000	1.028336e-12	1.057475e-24	550.992000	550.992000

Fig.18 **-** *No passenger scenario*

5.1.4 Limitations

Despite the model's comprehensiveness, several limitations persist. The simulation assumes average traffic flow conditions and does not yet incorporate real-time congestion data or stochastic delays, which may affect charging opportunities or energy use. Additionally, battery aging is modeled at a simplified aggregate level, without cell-level degradation modeling. Infrastructure constraints, such as limited charger availability or depot queue dynamics, are acknowledged but not explicitly simulated in queueing-theoretic terms.

5.1.5 The Sustainability of the Solution and Data Requirements

The solution is environmentally and operationally sustainable, with CO_2 emissions significantly reduced and battery usage managed within industry-recommended thresholds. Sustainability hinges on ongoing access to passenger demand data, real-time SOC monitoring, and proper fleet management integration. These can be achieved through IoT-based battery management systems and centralized scheduling dashboards.

The solution is robust across a range of operational scenarios, including variable weather, partial route obstructions, and load fluctuations. Its performance remains within acceptable bounds under stress-test conditions, and simulation outputs demonstrate minimal variance in key performance metrics under ±15% parameter variation. This robustness supports confident deployment at scale.

5.1.6 Summary of Verification &Validation Contributions

Thanks to a robust verification and validation approach, validation was ensured that the simulation results (used to identify the most efficient and sustainable bus routes) are rooted in a structurally sound and empirically credible model. The initial testing with the real-world 28T route enabled the project to expand our scope confidently to additional lines. This adherence to methodical model validation, as emphasized by Barlas (1996), ensures that route prioritization results are defensible and reliable for long-term transport planning

5.2 Key Findings

5.2.1 Scenario Analysis

5.2.1.1 Scenario Analysis for Different Seasons

In order to analyze the feasibility of a bus line's electrification, the analyses should be done under the most difficult conditions regarding battery consumption. In the literature, the battery consumption of an electric bus is significantly higher due to the heating systems' operations to heat the car during daily operations. In the simulation, results were in accordance with literature such that there is a statistically significant difference in average Energy Consumption (kWh) per simulation run (per route) between seasons for both BYD K8 and Karsan e-ATAK across all the routes in the dataset (Figure 19). The sample mean and 95% confidence intervals for the energy consumption metric can be found in Appendix Table 1. As metrics were calculated for both summer and winter scenarios (Appendix A, Tables 2 and 3), both types of electric buses showed significantly higher energy consumption levels during winter ($p<0.05$), which was expected due to heating and passenger distribution. Since CO_2 emissions are a correlate of the energy consumption in this study, it was also significantly higher during Winter compared to Summer. Since the feasibility of a bus line can best be measured under the most challenging conditions possible, the simulation results were analyzed for only the Winter season.

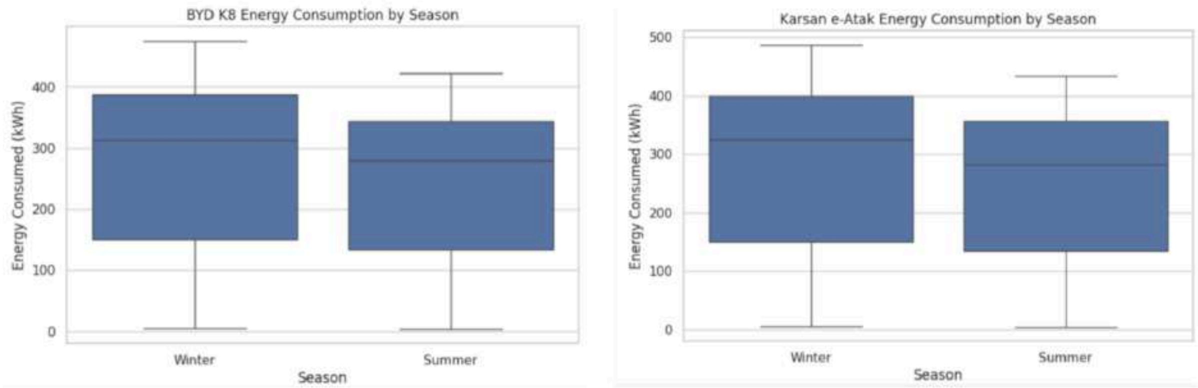


Fig.19- *Average energy consumption (kWh) over all routes for BYD K8 and Karsan e-Atak electric bus models in different seasons*

5.2.1.2 Scenario Analysis for Different Routes

The simulation is conducted for 13 different routes that have distinct parameters such as travel distance, max slope across the route, bus stop frequency, and passenger distribution. Each route is a unique environment for the buses to operate, and scenario-based analysis provides insight into which lines are feasible for electric bus operations. Therefore, each route is treated as a different scenario. Given the distinct characteristics of routes, significant differences in metrics such as operational feasibility and CO_2 emissions are expected prior to analysis.

__Table 9__- Route Characteristics

Line	First Stop	# of Stops	# of Passengers in February	Max Slope btw 2 stops
32	EMİNÖNÜ	39	101.403	-6,36
33	EMİNÖNÜ	51	102.647	9,53
35	EMİNÖNÜ	13	154.584	6,93
80	EMİNÖNÜ	17	2.261	5,21
90	EMİNÖNÜ	10	51.323	11,31
93	EMİNÖNÜ	39	154.646	8,03
336E	EMİNÖNÜ	44	405.916	5,21
33B	EMİNÖNÜ	54	106.437	-12,88
33E	EMİNÖNÜ	14	1.711	5,41
33ES	EMİNÖNÜ	30	7.850	16,67
33TE	EMİNÖNÜ	28	11.990	-8,39
33Y	EMİNÖNÜ	30	30.334	6,41
36KE	EMİNÖNÜ	37	65.688	-8,66
38E	EMİNÖNÜ	37	146.527	-10,73
99A	EMİNÖNÜ	17	164.590	16,67

__Table 10__- Route classification in terms of distance

Short Distance Routes	**Medium Distance Routes**	**Long Distance Routes**
90,33E,35,99A,80	33TE,33ES,33Y,36KE,38E,93,32,336E	33, 33B

Firstly, the impact of route length on average energy consumption was analyzed with electric buses. After comparison, the mean difference between the average energy consumption levels for each aggregated route class, provided in Table 10, was significant (Figure 20). However, since energy consumption is affected by the interaction of multiple factors such as distance, passenger distribution, and slope, bus routes exhibit unique operational characteristics, requiring route-level analysis. The route-level analysis will be discussed in detail in the section **Route Evaluation and Selection Process.**

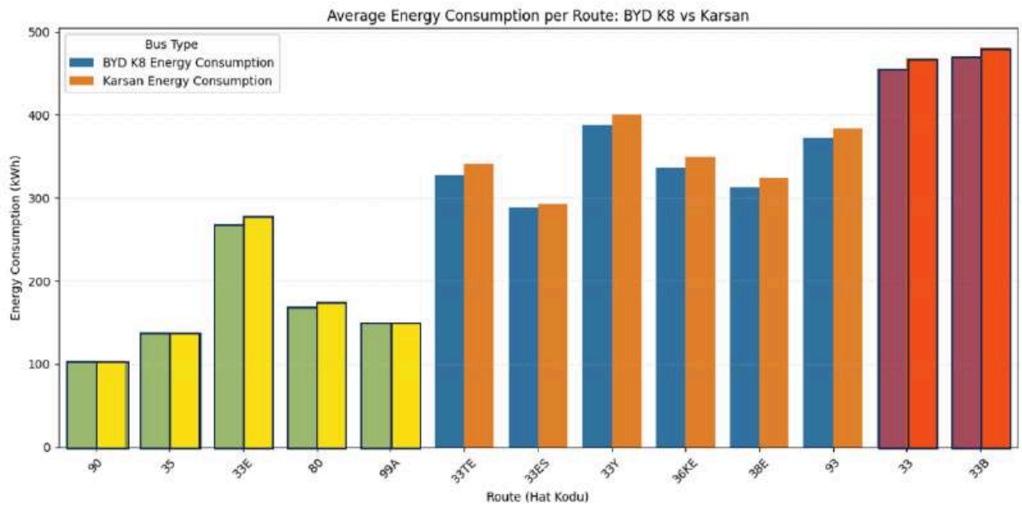


Fig.20 *- Average energy consumption (kWh) over all routes for BYD K8 and Karsan e-Atak electric bus models for different distance route classes*

5.2.2 Policy Analysis

5.2.2.1 Policy Analysis for Electric and Diesel Bus Models

For policy analysis, the average energy consumption and average CO_2 emissions will be analyzed for different bus models. By using two-sample t-tests, it was observed that the diesel buses' average CO_2 emissions for all routes were significantly higher compared to both BYD K8 and Karsan e-ATAK electric bus models during the Winter season (Fig. 21).

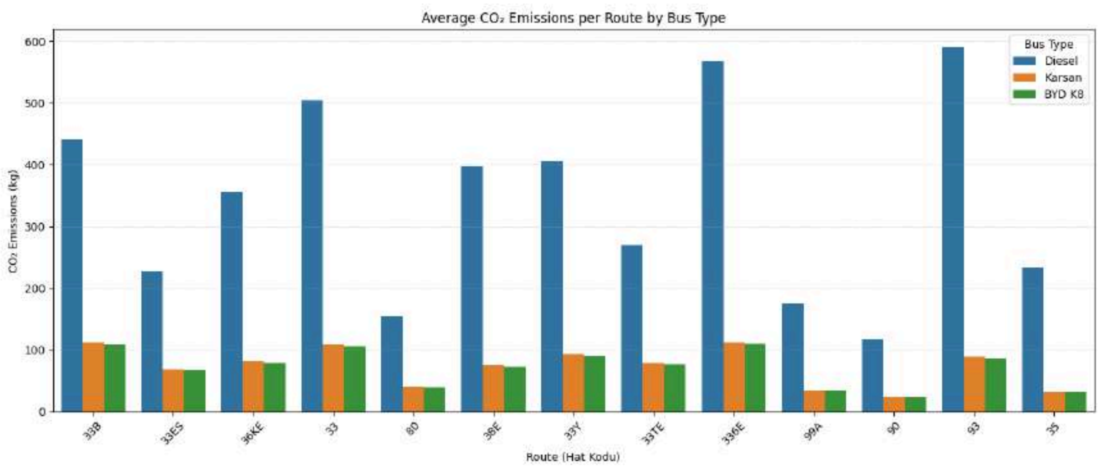


Fig.21- *Average CO_2 emissions results comparison for Diesel, BYD K8, and Karsan e-Atak electric bus models in Winter*

Since electric and diesel buses are powered by different energy sources, the units representing these sources are different. In order to compare the average energy consumption between an electric bus and a diesel bus, one must use a conversion factor to transform one unit into another. In this study, diesel buses' energy consumption measured in liters is converted to kWh by using the 9.8 kwH/L conversion factor in Table 2. The average energy consumption difference (kWh) per simulation run (per route) between BYD K8 and Karsan ATAK (diesel) was statistically significant ($p<0.05$) across all the routes in the dataset (Fig. 22) with BYD K8 having significantly lower average energy consumption compared to the diesel bus. Electric buses have superior regenerative energy dynamics compared to diesel buses, decreasing the overall energy consumption.

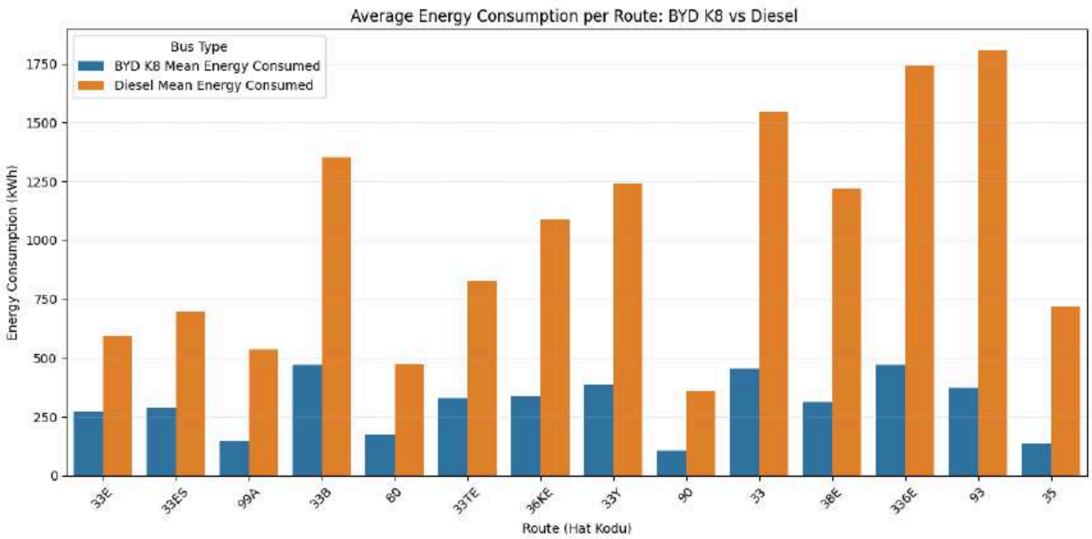
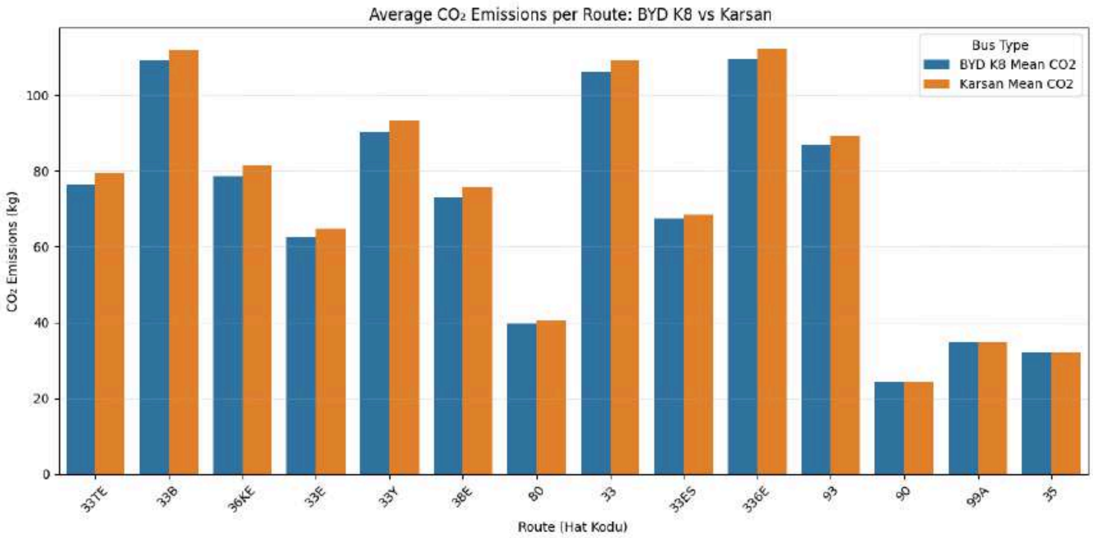


Fig.22 - *Average energy consumption results for BYD K8 and Karsan ATAK bus models*

5.2.2.2 Policy Analysis for Different Electric Bus Models

There is a statistically significant difference in average CO_2 emissions per simulation run (per route) between BYD K8 and Karsan e-ATAK across all the routes in the dataset except for routes 90, 99A, and 35 which can be seen in Figure 23. BYD K8 has significantly lower average CO_2 emissions compared to Karsan e-Atak in all the routes except for 90, 99A,



and 35.

Fig.23 - *CO_2 emissions results for BYD K8 and Karsan e-Atak electric bus models*

The average energy consumption of the Karsan e-ATAK bus model is significantly higher than the BYD K8 for all the routes except for 90, 99A, and 35 (Fig. 24). The BYD K8 bus model has a larger battery compared to Karsan e-ATAK, therefore it had fewer depot visits for charging. Fewer visits to the depots for charging correlate to smooth bus operations with fewer delays thus, less energy consumption.

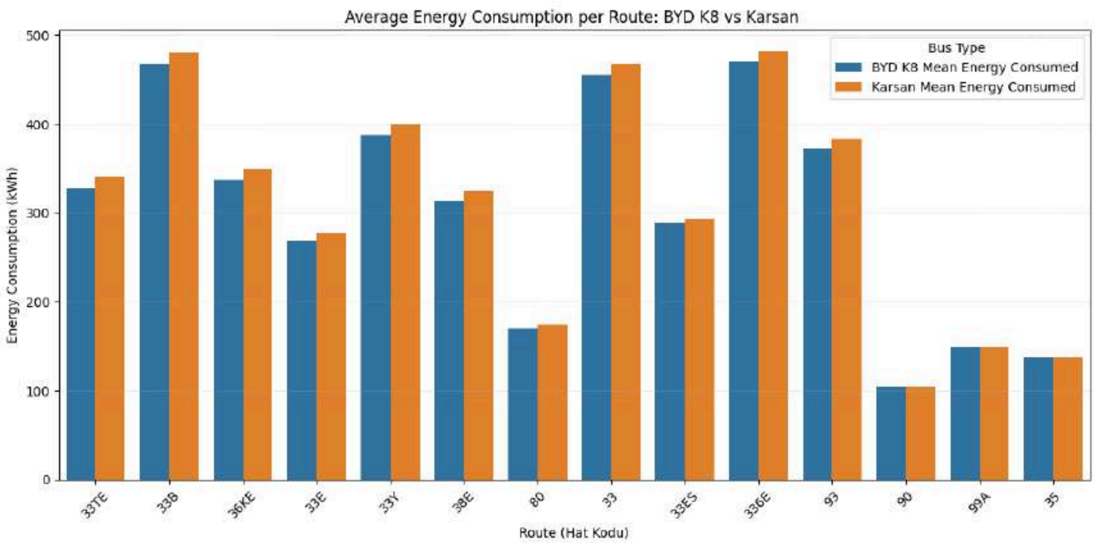


Fig.24 - *Average energy consumption between Karsan e-ATAK and BYD K8 electric bus models*

5.2.3 Route Evaluation and Selection Process

The evaluation of routes for the integration of electric buses into the Istanbul public transport system was carried out systematically based on different criteria.

Schedule: Firstly, the suitability of each line found using the dataset provided by IETT to the daily operational programs according to the number of trips per vehicle was tested. Line 33Y was not suitable for e-bus transition because it could not complete the planned trips with both BYD and Karsan buses within the specified time period (06:00–00:00). Meanwhile, 336E and 33B completed the program with BYD buses but not with Karsan. Since Karsan buses are not applicable on these two routes, it is possible that BYD K8 is also not applicable under different scenarios. Therefore, 336E and 33B are marked as questionable choices in terms of applicability.

Minimum Battery Level & Battery Life: In the second stage, minimum battery level analysis was performed on simulation outputs. The objective was not to let the battery charge rate fall below 15%, and lines exceeding this limit were accepted for sustainability as detailed in the literature review. Minimum state of charge before recharging for the bus on each route across 100 simulation replications are shown in Figure 25 for BYD K8 and Figure 26 for Karsan.

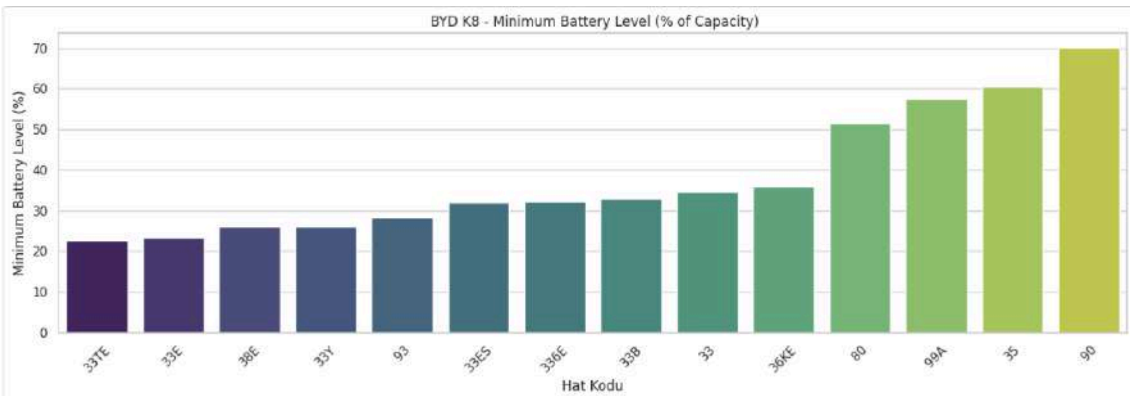


Fig.25 *- Minimum battery level before recharging for BYD K8*

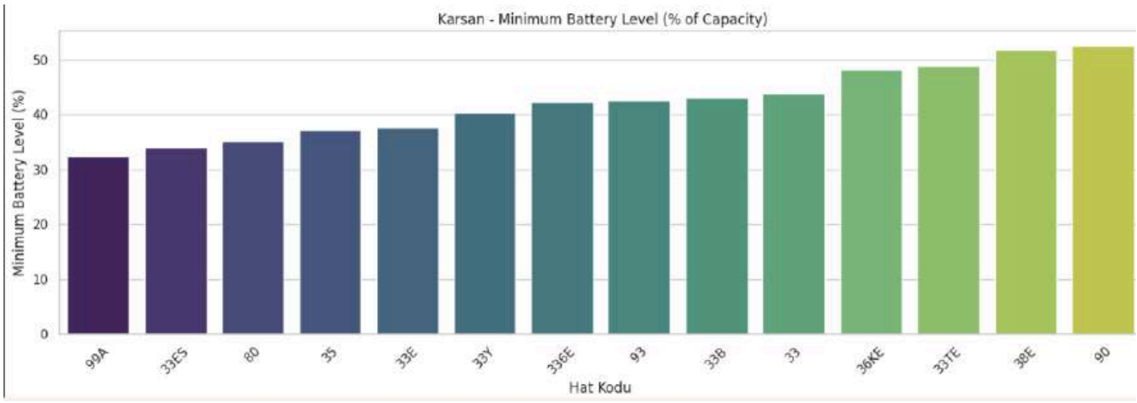


Fig.26 - *Minimum battery level before recharging for Karsan*

As can be seen from the plots, no line falls below the 15% limit. However, in the BYD plot, lines 33TE, 33E, 38E, and 33Y are seen to be close to the limit. In real life conditions, unmodeled factors can make these lines more challenging. Therefore, these lines are marked as questionable choices and will be reconsidered during route selection.

Total Passengers Traveled & Utility: In the simulation, the routes are also analyzed in terms of the total number of passengers traveled, and as a result, the total number of passengers traveled aligns well with the real-world data, indicating the accuracy of the simulation, but none of the lines can be eliminated in this step. Figures 27 and 28 show the utilization levels for BYD and Karsan buses. When the two plots are compared, it is seen that there are differences in the utilization levels, even though the same seed has been used in the simulation. This is because the buses have different capacities. Since Karsan has a smaller capacity, it reaches a higher utilization level faster.

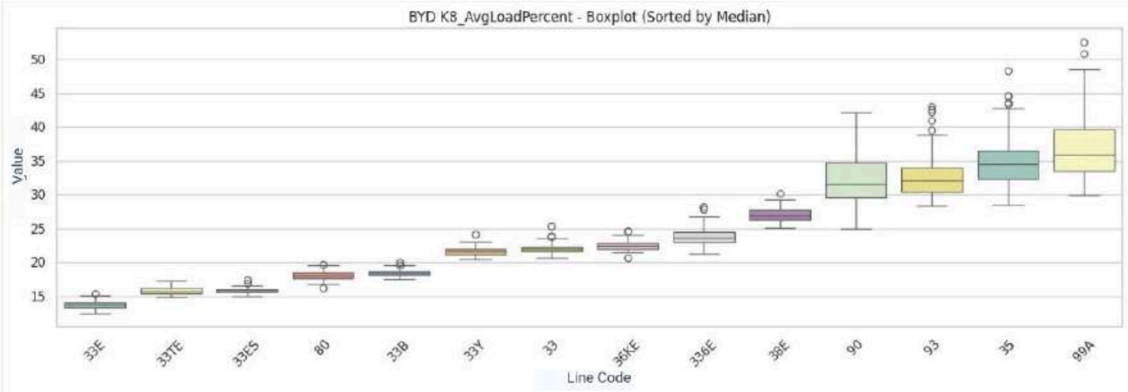


Fig.27 *- Utilization levels of the BYD buses*

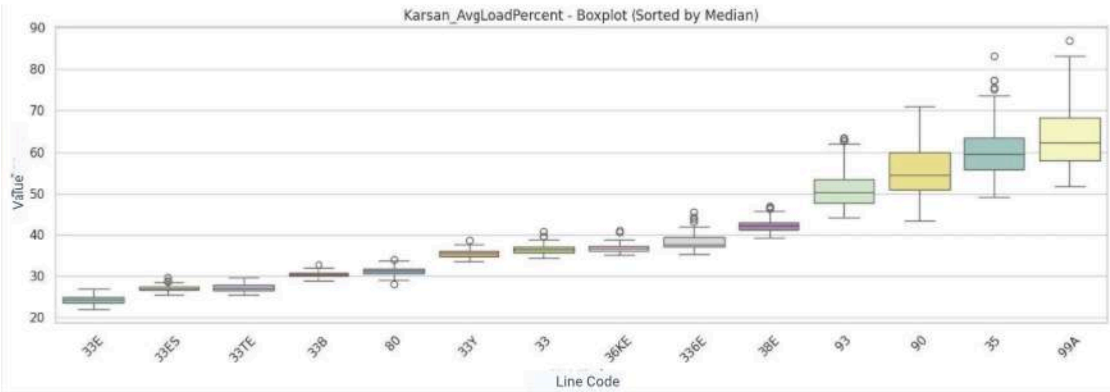


Fig.28 *- Utilization levels of the Karsan buses*

Total Energy Consumption: Energy consumption of diesel and electric buses was compared. Analyses were made assuming that 1 liter of diesel is equivalent to 9.8-10 kWh of energy as detailed in the literature review. At this stage, none of the routes could be eliminated based on the energy consumption but line 93 was determined as the priority candidate due to its highest energy consumption in the diesel bus and CO_2 emission reduction potential. Line-based plots for BYD and diesel buses are given in Figures 29 and 30. Since the consumption plots of BYD and Karsan buses are similar, a separate Karsan plot has not been added in this report.

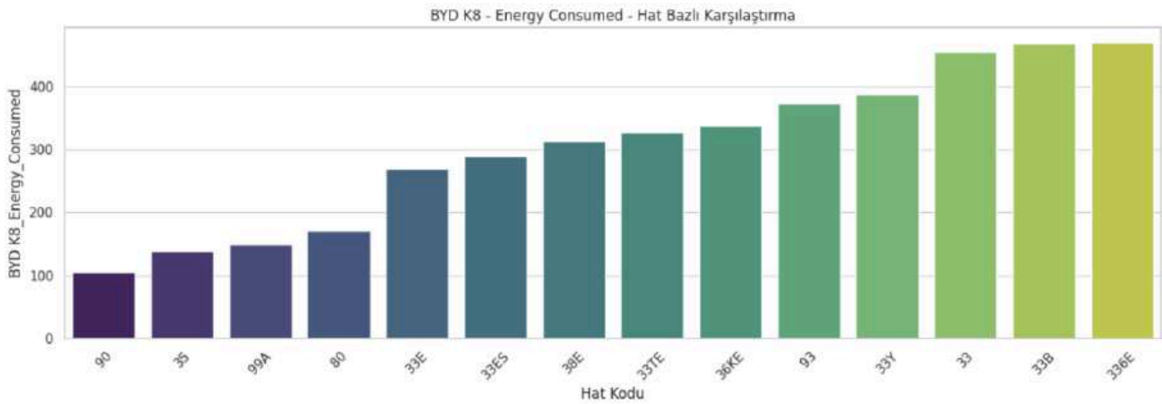


Fig.29 - *Energy consumptions of the BYD buses*

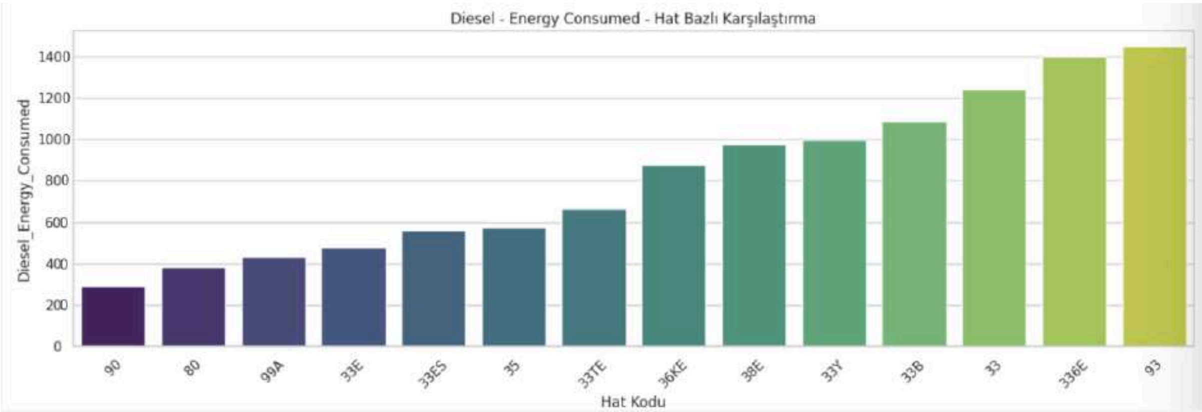


Fig.30- *Energy consumptions of the diesel buses*

Total CO2 Emission: In the electric bus integration process, the CO_2 emission performances of diesel and electric vehicles were comprehensively compared. The analyses revealed that e-buses exhibited significantly lower emission values compared to diesel buses on all lines as shown in Figure 21. Simulation data show that even under the same operational conditions, electric buses provide a clear superiority in terms of emissions.

In order to quantitatively evaluate the emission reduction potential, the average CO_2 differences calculated in kg for each line are presented in Table 11. Line 93 stands out as having the highest reduction potential with an emission difference of 387.6 kg. Lines 336E and 33 follow with values of 380.8 kg and 379.4 kg, respectively whereas line 90 showed the lowest performance with 59 kg.

Table 11 - CO_2 emission differences between diesel and e-buses

Code	Difference (kg)
93	387,6
336E	380,8
33	379,4
36KE	286,2
38E	268,3
33B	254,7
33Y	233,2
35	180,3
33TE	176,1
33E	152,9
33ES	140,6
99A	125,2
80	122,3
90	59

Based on the decision metrics defined above, in the initial stage of selecting the routes to be electrified, it is ensured that the schedule of the selected route can be completed by both electric bus types, given that operational parameters such as driver behavior, travel speed, boarding and alighting times, and 18-hour shifts are defined within the simulation framework based on simplifying assumptions that may not fully reflect real-world conditions. In real-world operations, these parameters may exhibit greater variability than anticipated. Moreover, exogenous factors that were not explicitly modeled in the simulation may result in schedule infeasibility due to their influence on the energy consumption. To mitigate such operational risks, the lines that demonstrated schedule feasibility for both e-bus types under the defined simulation parameters were prioritized in the first stage.

When selecting among the remaining routes, those with the highest potential for CO_2 emission reduction compared to diesel vehicles were prioritized for electrification, in order to minimize environmental impact. Accordingly, the first routes to be electrified are determined to be 93, 33, 36KE, and 35, respectively.

In addition to the criteria used in the first stage, the average charge count was also considered in the selection process for the second stage. Since real-world depot charging facilities may be limited, frequent returns to the depot could pose additional challenges for completing the schedule. However, the analysis of average charge counts showed no significant differences among the routes. Therefore, the decision for the second-stage transition process was primarily based on the potential for CO_2 reduction. In this case, the routes selected for electrification in the second stage were determined to be 336E, 38E, and 33B, respectively.

As a result, by excluding the lines marked as questionable by looking at the previous parameters, the lines with high CO2 reduction were taken as first stage prioritization lines which are 93, 33, 36KE and 35.

For the second stage of the ranking, the same logic is applied to the remaining routes. Considering the significant emissions savings, routes marked as questionable are reincluded and second stage prioritization lines are determined as 336E, 38E and 33B.

6. Suggestions for a Successful Implementation

The decision support tool developed in this study uses a modular approach where the system's main components -such as passenger distributions and route characteristics- can easily be updated with the user's input. This modular design provides flexibility and future scalability to the simulation model. IETT can utilize this tool with the following steps.

Firstly, the implementation of the solution requires a pilot region of routes. After determining the pilot region, IETT should provide the relevant data including technical bus properties with geographic route characteristics(e.g. slope and distance) to the simulation. The simulation will process the provided data and calculate the operational and environmental performance metrics. After obtaining the simulation results, a route-level analysis should be made based on the decision criteria aligned with IETT's goals.

The design can be integrated with the overall bus public transportation system due to its modular nature. Should IETT want to extend the route electrification project's scope, it only needs to provide the data regarding the potential routes' characteristics such as travel distances between each route segment and passenger distributions. The simulation does not require rebuilding, making it flexible for experimenting with alternative subsets of routes and buses. Due to the rapid technological developments regarding battery capacities and charging infrastructure, the method should be revised at least annually to reflect these developments for performance metric calculations' accuracy.

7. Conclusions and Discussion

7.1 Summary of Industrial Engineering Methods Integrated for the Project

Data analysis and simulation modeling methods were used in the design and implementation process of the decision-support tool developed for IETT. During implementation, a discrete-event simulation was built on SimPy. Statistical methods were used to assess the statistical reliability of the simulation output by calculating sample means and building 95% confidence intervals on the results obtained from 100 replications under different scenarios. Finally, route-level analysis was conducted to determine the feasible routes for electric bus conversion. The routes with high CO_2 emission reduction were determined as first-stage prioritization lines which are 93, 33, 36KE, and 35. The second stage prioritization lines are determined as 336E, 38E, and 33B.

7.2 Ecological and Societal Significance of the Design

The decision-aid tool developed in this project will support IETT's goals for reducing greenhouse gas emissions by choosing the routes that would be the most feasible to be electrified within a chosen region such as the Historical Peninsula. In the light of comparison between electric and diesel buses' CO_2 emissions in Table A, if IETT deploys any of the proposed electric bus models on the candidate routes, the CO_2 emissions will decrease significantly within the Historical Peninsula. Deploying electric buses suggested by this study will provide further progress for IETT in its sustainability goals.

If a selection among the proposed electric bus models BYD K8 and Karsan e-Atak is required, their performance metrics must be compared. Among the candidate electric bus models, the BYD K8 provided significantly lower CO_2 emissions compared to the Karsan e-ATAK across the majority of the routes. Due to its large battery capacity, the BYD K8 model had fewer depot visits for charging compared to the Karsan e-ATAK, and it carried more passengers throughout its operations due to its higher passenger capacity. By using BYD K8 electric bus models, IETT can decrease the average CO_2 emissions caused by public transportation in the chosen routes, bringing it closer to its sustainability goals (Fig. 31). Additionally, BYD K8 electric bus models had a significantly lower energy consumption across all the routes than diesel buses (Fig. 32). In conclusion, this study suggests prioritizing the deployment of the BYD K8 model on the chosen routes due to its better performance in terms of operational efficiency and environmental impact. When IETT deploys BYD K8 on the chosen routes for the first stage of electrification, it can benefit from environmental and economic benefits due to lower average energy consumption and CO_2 emissions compared to Karsan e-ATAK and Karsan ATAK.

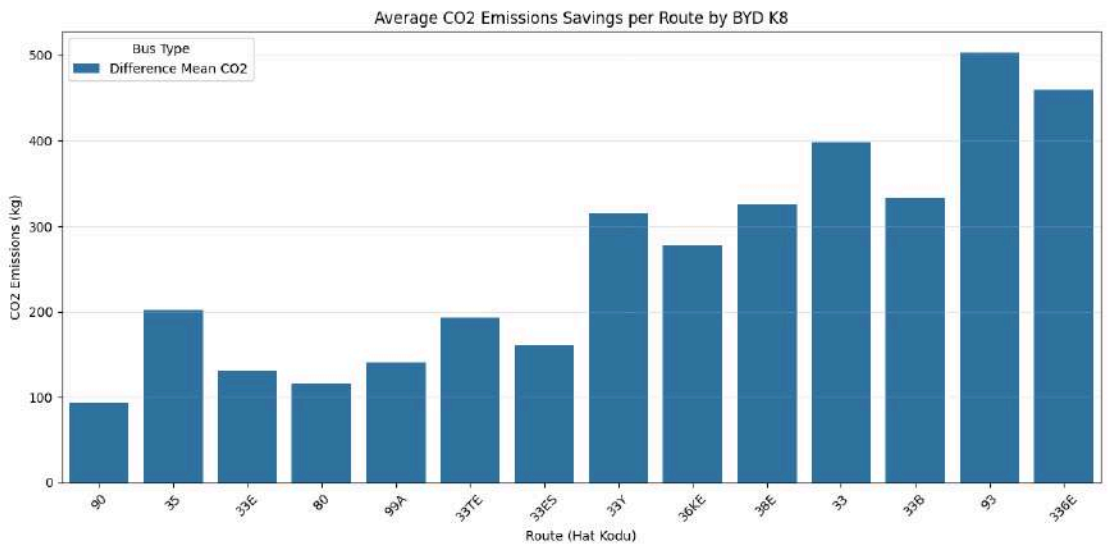


Fig.31- *Average CO2 emissions saved per route by BYD K8*

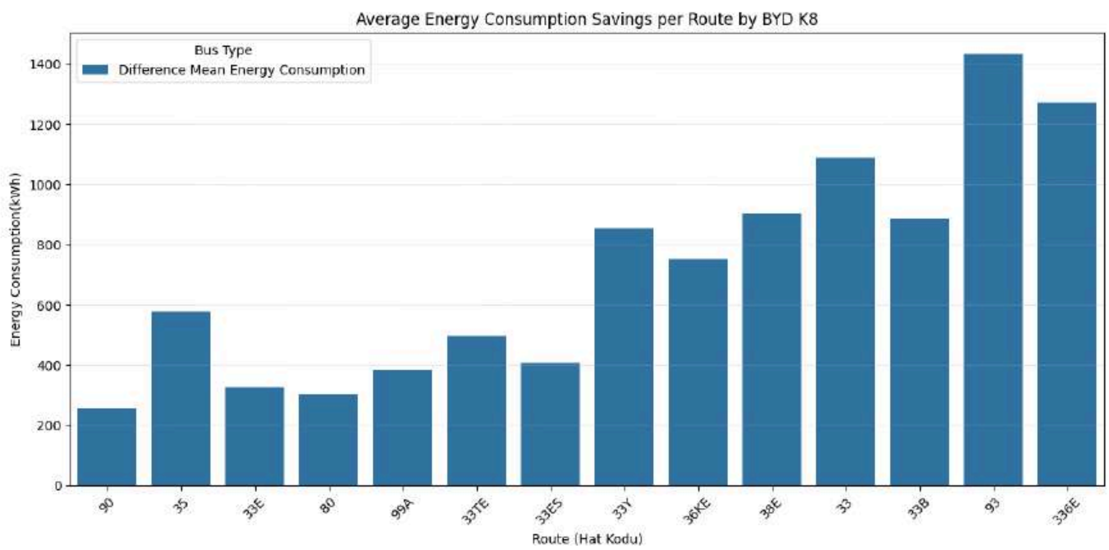


Fig.32- *Average energy consumption savings per route by BYD K8*

Utilizing electric buses on the chosen routes will also provide societal impacts by reducing the noise pollution and increasing the passenger satisfaction. Reducing the noise caused by the bus engines will help create an overall quiet public space. Due to electric buses' lower vibration, passenger satisfaction for ride experience will increase as well compared to traditional diesel buses.

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Appendix A

Table 1: *Energy consumption results for electric and diesel bus models*

	BYD K8			Karsan e-ATAK			Karsan ATAK		
Route Code	**Mean (kwH**	**95% Confidence Interval Lower**	**95% Confidence Interval Upper**	**Mean (kWh**	**95% Confidence Interval Lower**	**95% Confidence Interval Upper**	**Mean (lt)**	**95% Confidence Interval Lower**	**95% Confidence Interval Upper**
33	455,491707	455,287789	455,695625	467,880898	467,694967	468,066829	157,702894	156,370112	159,035676
35	138,119331	137,844037	138,394624	138,113519	137,841865	138,385172	73,052208	71,0968824	75,0075336
80	170,268795	170,202929	170,33466	174,747973	174,680663	174,815283	48,5131009	48,1263192	48,8998826
90	104,405684	104,171967	104,639402	104,340093	104,126357	104,553828	36,8956233	36,3375672	37,4536793
93	372,735533	372,007936	373,46313	383,291307	382,696577	383,886037	184,250655	180,800233	187,701078
336E	469,948727	469,613628	470,283826	481,235667	480,943054	481,528281	177,751339	175,865862	179,636815
33B	467,993126	467,861656	468,124596	480,345175	480,220641	480,469708	138,094476	137,273184	138,915767
33E	268,544635	268,447344	268,641926	277,746189	277,648206	277,844172	60,5386067	60,0699319	61,0072814
33ES	288,912271	288,838555	288,985986	293,284362	293,212858	293,355866	70,9614789	70,5102583	71,4126995
33TE	327,604996	327,501028	327,708963	340,984083	340,865622	341,102545	84,2578101	83,5239614	84,9916589
33Y	387,922586	387,749355	388,095817	400,196518	400,039256	400,35378	126,853838	125,786633	127,921044
36KE	337,158795	337,025404	337,292186	349,586735	349,460819	349,712651	111,220002	110,422456	112,017547
38E	313,115798	312,94355	313,288045	324,469456	324,306416	324,632496	124,24529	123,15692	125,33366
99A	148,879751	148,51318	149,246323	148,812265	148,47439	149,15014	54,6211695	53,9521285	55,2902105

Table 2*: Average Energy Consumption (kWh), Average CO_2 Emissions Difference (kg), Average Recharge Count, Schedule Completion Metrics for all routes for different E-Bus types for Summer*

	SUMMER													
	33		33B		33E		33ES		33TE		33Y		336E	
	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN
Average Energy Consumption	407,8	418,65	416,44	427,85	242,28	250,48	254,98	263,02	294,24	306,18	345,31	357,06	419,39	430,67
Average CO2 Emissions Difference	379,42	376,88	254,71	252,05	152,97	151,06	140,69	138,83	176,08	173,29	233,27	230,53	380,86	378,23
Average Recharge Count	2	5	2	2	0	2	0	2	1	4	1	4	2	5
Is the Schedule Completed?	+	+	+	+	+	+	+	+	+	+			+	+

	SUMMER													
	35		36KE		38E		80		90		93		99A	
	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN
Average Energy Consumption	122,77	122,75	302,53	313,33	279,21	282,55	153,36	153,65	90,59	90,59	328,86	338,79	133,05	132,99
Average CO2 Emissions Difference	180,31	180,31	286,22	283,7	268,31	267,53	122,32	122,26	59,01	59,02	387,69(	385,38	125,2	125,22
Average Recharge Count	2	0	1	4	1	2	0	0	0	0	1	4	0	0
Is the Schedule Completed?			+	+	+	+	+	+			+	+		

***Table 3**: Average Energy Consumption (kWh), Average CO_2 Emissions Difference (kg), Average Recharge Count, Schedule Completion Metrics for all routes for different E-Bus types for Winter*

WINTER	33		33B		33E		33ES		33TE		33Y		336E	
	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN
Average Energy Consumption	455,49	467,88	467,99	480,34	268,54	277,74	288,91	293,28	327,6	340,98	387,92	400,19	469,95	481,23
Average CO2 Emissions Difference	398,52	395,63	332,86	329,98	131,15	129,01	159,76	168,74	193,29	190,18	315,55	312,68	459,3	456,67
Average Recharge Count	2	5	2	5	0	2	1	2	1	4	1	4	2	5
Is the Schedule Complete?	+	+	+	+	+	+	+	+	+	+			+	+

WINTER	35		36KE		38E		80		90		93		99A	
	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN	BYD K8	KARSAN
Average Energy Consumption	138,12	138,11	337,16	349,58	313,11	324,46	170,27	174,75	104,4	104,34	372,73	383,29	148,88	148,81
Average CO2 Emissions Difference	201,59	201,58	277,35	274,45	324,63	321,98	115,57	114,52	93,74	93,75	502,75	500,29	140,1	140,11
Average Recharge Count	0	0	1	4	1	4	0	1	0	0	1	4	0	0
Is the Schedule Complete?			+	+	+	+	+	+			+	+		

```
T-test and F-value results for 'BYD K8_Energy_Consumed'

Short vs Medium:
  t-statistic = -58.500, p-value = 0.00e+00, F-value = 1.010

Short vs Long:
  t-statistic = -117.118, p-value = 0.00e+00, F-value = 77.122

Medium vs Long:
  t-statistic = -48.651, p-value = 2.48e-235, F-value = 77.908
```

***Fig. 1** : t-test Results for average energy consumption for route classes for BYD K8 (Fig. 21)*

```
Hat Kodu  BYD K8 Mean Energy Consumed  Diesel Mean Energy Consumed  \
     33E                   268.544635                   593.278345
    33ES                   288.912271                   695.422493
     99A                   148.879751                   535.287461
     33B                   467.993126                  1353.325861
      80                   170.268794                   475.428389
    33TE                   327.604996                   825.726539
    36KE                   337.158795                  1089.956015
     33Y                   387.922586                  1243.167614
      90                   104.405684                   361.577108
      33                   455.491707                  1545.488361
     38E                   313.115798                  1217.603842
    336E                   469.948727                  1741.963117
      93                   372.735533                  1805.656422
      35                   138.119331                   715.911638

 T-statistic        P-value  Diesel Higher?  Significant (p < 0.05)
  851.724343  1.083475e-207            True                    True
  934.342474  2.251293e-206            True                    True
  242.169266  3.275951e-200            True                    True
  777.400808  1.071254e-197            True                    True
  608.235641  1.198618e-187            True                    True
  643.517765  2.050633e-187            True                    True
  547.646636  1.435629e-182            True                    True
  473.277117  1.761966e-175            True                    True
  218.702683  1.325327e-171            True                    True
  432.893050  1.263211e-170            True                    True
  335.947885  1.208722e-159            True                    True
  299.060135  5.147641e-156            True                    True
  104.764215  2.801261e-110            True                    True
   64.585695   4.141394e-85            True                    True
```

***Fig.2** - t-test results for average energy consumption between diesel and BYD K8 bus models (Fig. 22)*

Hat Kodu	BYD K8 Mean CO2 Emissions	Karsan Mean CO2 Emissions	\
33TE	76.331964	79.449291	
33B	109.042398	111.920426	
36KE	78.557999	81.453709	
33E	62.570900	64.714862	
33Y	90.385963	93.245789	
38E	72.955981	75.601383	
80	39.672629	40.716278	
33	106.129568	109.016249	
33ES	67.316559	68.335256	
336E	109.498053	112.127911	
93	86.847379	89.306875	
90	24.326524	24.311242	
99A	34.688982	34.673258	
35	32.181804	32.180450	

T-statistic	P-value	Karsan Higher?	Significant (p < 0.05)
-166.374239	7.387036e-212	True	True
-133.691935	1.338777e-195	True	True
-132.793407	5.709950e-195	True	True
-130.612832	4.694356e-194	True	True
-102.822024	1.360050e-172	True	True
-93.826416	1.043230e-165	True	True
-93.222081	1.785058e-165	True	True
-87.994515	1.060928e-159	True	True
-83.442226	3.920496e-156	True	True
-49.727222	1.572614e-112	True	True
-22.016124	3.112546e-54	True	True
0.405919	6.852442e-01	False	False
0.265326	7.910362e-01	False	False
0.029453	9.765326e-01	False	False

Fig.3 - *t-test results for average CO2 between Karsan e-ATAK and BYD K8 (Fig. 23)*

Hat Kodu	BYD K8 Mean Energy Consumed	Karsan Mean Energy Consumed	\
33TE	327.604996	340.984083	
33B	467.993126	480.345175	
36KE	337.158795	349.586735	
33E	268.544635	277.746189	
33Y	387.922586	400.196518	
38E	313.115798	324.469456	
80	170.268794	174.747973	
33	455.491707	467.880898	
33ES	288.912271	293.284362	
336E	469.948727	481.235667	
93	372.735533	383.291307	
90	104.405684	104.340092	
99A	148.879751	148.812265	
35	138.119331	138.113519	

T-statistic	P-value	Karsan Higher?	Significant (p < 0.05)
-166.374239	7.387036e-212	True	True
-133.691935	1.338777e-195	True	True
-132.793407	5.709950e-195	True	True
-130.612832	4.694356e-194	True	True
-102.822024	1.360050e-172	True	True
-93.826416	1.043230e-165	True	True
-93.222081	1.785058e-165	True	True
-87.994515	1.060928e-159	True	True
-83.442226	3.920496e-156	True	True
-49.727222	1.572614e-112	True	True
-22.016124	3.112546e-54	True	True
0.405919	6.852442e-01	False	False
0.265326	7.910362e-01	False	False
0.029453	9.765326e-01	False	False

Fig.4*: t-test results for average energy consumption between Karsan e-ATAK and BYD K8 electric bus models (Fig. 24)*

Hat Kodu	Difference Mean CO2	T-statistic	P-value	Diesel Higher?	\
33B	4221.600358	-248.222254	3.402130e-140	True	
33ES	2158.035419	-216.847796	2.144663e-134	True	
36KE	3409.301248	-212.980693	1.265019e-133	True	
33	4839.433188	-183.134357	3.838709e-127	True	
80	1481.698215	-182.999219	4.070736e-127	True	
38E	3823.376313	-182.678767	4.888636e-127	True	
33Y	3887.750403	-181.088490	1.155239e-126	True	
33E	1835.919805	-171.380736	2.551821e-124	True	
33TE	2565.992962	-161.321642	1.055809e-121	True	
336E	5464.783920	-149.193896	2.286810e-118	True	
99A	1678.230894	-128.156993	4.072218e-112	True	
90	1132.720222	-102.836893	1.374082e-102	True	
93	5691.253170	-89.235732	1.931762e-96	True	
35	2258.735439	-63.142673	8.327012e-82	True	

Fig.5 - *t-test results for average CO2 emissions saved per route by BYD K8 (Fig.31)*

Hat Kodu	T-statistic	P-value	Diesel Higher?
33E	851.724343	1.083475e-207	True
33ES	934.342474	2.251293e-206	True
99A	242.169266	3.275951e-200	True
33B	777.400808	1.071254e-197	True
80	608.235641	1.198618e-187	True
33TE	643.517765	2.050633e-187	True
36KE	547.646636	1.435629e-182	True
33Y	473.277117	1.761966e-175	True
90	218.702683	1.325327e-171	True
33	432.893050	1.263211e-170	True
38E	335.947885	1.208722e-159	True
336E	299.060135	5.147641e-156	True
93	104.764215	2.801261e-110	True
35	64.585695	4.141394e-85	True

Fig.6 - *t-test results for average energy consumption savings per route by BYD K8 (Fig.32)*

Appendix B:

Screenshots of Main Parts of the Simulation

Classes:

Bus:

```
class Bus:
    def __init__(self, env, name, route_segments, depot_to_start, rounds, capacity, battery_capacity, co2_per_kwh, bus_area=22):
        self.env = env
        self.name = name
        self.route_segments = route_segments
        self.depot_to_start = depot_to_start
        self.rounds = rounds
        self.capacity = capacity
        self.bus_area = bus_area

        self.passengers = 0
        self.battery_capacity = battery_capacity
        self.battery_level = battery_capacity
        self.co2_per_kwh = co2_per_kwh

        self.total_co2_emissions = 0
        self.total_energy_consumed = 0
        self.passenger_counts = []
        self.routes_completed = 0
        self.total_passenger_transported = 0
        self.end_of_service_time = None

        self.action = env.process(self.run())
```

Fig.7 - *Attributes of bus class*

Functions:

```python
def slope_correction(self, slope):
    return 1 + (slope / 10) if slope > 0 else max(0.7, 1 + (slope / 25))

def consumption_rate(self, passenger_mass, slope):
    base = np.interp(passenger_mass, [0, 1000, 2000, 4000, 5700], [0.8, 0.9, 1.0, 1.2, 1.4])
    return max(base * self.slope_correction(slope), 0.6)

def travel_segment(self, segment, passenger_mass):
    yield self.env.timeout((segment.distance_km / 30) * 60)

    leaving = random.randint(0, min(self.passengers, 10))
    self.passengers -= leaving
    boarding = min(random.randint(5, 20), self.capacity - self.passengers)
    self.passengers += boarding

    passenger_mass = self.passengers * 70
    self.passenger_counts.append(self.passengers)
    self.total_passenger_transported += boarding

    density = self.passengers / self.bus_area
    boarding_time = boarding * np.interp(density, [0, 1, 2, 4, 6], [0.89, 1.13, 1.37, 1.67, 2.03])
    alighting_time = leaving * np.interp(density, [0, 1, 2, 4, 6], [0.59, 0.92, 1.11, 1.89, 5.92])
    wait_time = max(boarding_time, alighting_time) + 8.8
    self.total_stop_time += wait_time / 60
    yield self.env.timeout(wait_time / 60)

    consumption = self.consumption_rate(passenger_mass, segment.slope_percent) * segment.distance_km
    self.battery_level -= consumption
    self.total_energy_consumed += consumption
    self.total_co2_emissions += consumption * self.co2_per_kwh
    self.min_battery_level = min(self.min_battery_level, self.battery_level)

def charge(self):
    to_charge = self.battery_capacity * 0.8 - self.battery_level
    self.total_energy_charged += to_charge
    self.charge_count += 1
    yield self.env.timeout(60)
    self.battery_level = self.battery_capacity * 0.8

def run(self):
    yield from self.travel_segment(self.depot_to_start, 0)
    for _ in range(self.rounds):
        route_distance = sum(s.distance_km for s in self.route_segments) * 2
        avg_mass = np.mean(self.passenger_counts) * 70 if self.passenger_counts else 0
        est_need = self.consumption_rate(avg_mass, 0) * route_distance + 50

        if self.battery_level < est_need:
            yield from self.travel_segment(RouteSegment(self.depot_to_start.distance_km, -self.depot_to_start.slope_percent), 0)
            yield from self.charge()
            yield from self.travel_segment(self.depot_to_start, 0)

        for seg in self.route_segments:
            yield from self.travel_segment(seg, self.passengers * 70)
        for seg in reversed(self.route_segments):
            yield from self.travel_segment(RouteSegment(seg.distance_km, -seg.slope_percent), self.passengers * 70)

        self.passengers = 0
        self.routes_completed += 1

    yield from self.travel_segment(RouteSegment(self.depot_to_start.distance_km, -self.depot_to_start.slope_percent), 0)
    self.end_of_service_time = self.env.now
```

Fig.8 - *Functions of the buss class (Functions under Methodology and Assumptions topic)*

Simulation:

```python
class Simulation:
    def __init__(self, seed, route_segments, depot_distance_km=2.1, depot_slope=-0.5):
        self.seed = seed
        self.route_segments = route_segments
        self.env = simpy.Environment(initial_time=360)
        random.seed(seed)
        np.random.seed(seed)

        self.depot_to_start = RouteSegment(
            from_stop=-1,
            to_stop=0,
            distance_km=depot_distance_km,
            slope_percent=depot_slope,
            from_name="Edirnekapı Garajı",
            to_name="Topkapı"
        )

        self.bus1 = Bus(self.env, "BYD K8", self.route_segments, self.depot_to_start, 6, 90, 350, 0.233, 22)
        self.bus2 = Bus(self.env, "Karsan e-Atak", self.route_segments, self.depot_to_start, 6, 52, 220, 0.233, 17)

    def run(self, until=10000):
        self.env.run(until=until)
        total_stops = len(self.route_segments) * 2 * 6  # gidiş + dönüş * 6 tur

        return {
            "BYD K8_Passengers": self.bus1.total_passenger_transported,
            "BYD K8_Energy_Consumed": self.bus1.total_energy_consumed,
            "BYD K8_CO2": self.bus1.total_co2_emissions,
            "BYD K8_EndTime": format_time(self.bus1.end_of_service_time),
            "BYD K8_TotalServiceTime": self.bus1.end_of_service_time - 360 if self.bus1.end_of_service_time else None,
            "BYD K8_AvgStopTime": self.bus1.total_stop_time / total_stops if total_stops else None,
            "BYD K8_EnergyPerPassenger": self.bus1.total_energy_consumed / self.bus1.total_passenger_transported if self.bus1.total_passenger_transported else None,
            "BYD K8_ChargeCount": self.bus1.charge_count,
            "BYD K8_MinBatteryLevel": self.bus1.min_battery_level,

            "Karsan_Passengers": self.bus2.total_passenger_transported,
            "Karsan_Energy_Consumed": self.bus2.total_energy_consumed,
            "Karsan_CO2": self.bus2.total_co2_emissions,
            "Karsan_EndTime": format_time(self.bus2.end_of_service_time),
            "Karsan_TotalServiceTime": self.bus2.end_of_service_time - 360 if self.bus2.end_of_service_time else None,
            "Karsan_AvgStopTime": self.bus2.total_stop_time / total_stops if total_stops else None,
            "Karsan_EnergyPerPassenger": self.bus2.total_energy_consumed / self.bus2.total_passenger_transported if self.bus2.total_passenger_transported else None,
            "Karsan_ChargeCount": self.bus2.charge_count,
            "Karsan_MinBatteryLevel": self.bus2.min_battery_level,
        }
```

Fig.9 *- Simulation run*